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"Discipline is doing what you hate to do but doing it like you love it". --Mike Tyson

Introduction



For Har Gobind Khorana, the painstaking process of synthesizing short DNA sequences may have seemed slow and tedious. But his disciplined efforts helped pave the foundation for a medical revolution.

In the mid-1950s, in the laboratory of Sir Alexander Todd, Har Gobind Khorana pioneered the use of synthetic DNA and RNA to address fundamental biological questions. He demonstrated that these synthetic molecules could serve as precise templates for DNA and RNA polymerases and developed a technique that would not only help decipher the

genetic code but also established a powerful experimental framework for modern synthetic biology.

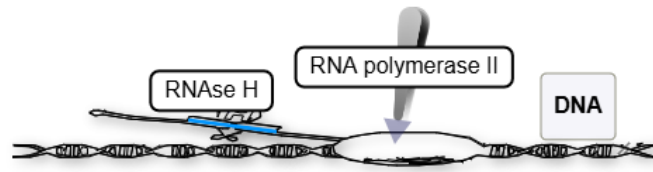
In the 1960s, while working with a postdoctoral associate, Dr. V. Mahadevan, Dr. Bob Letsinger developed the first method for synthesizing DNA on an insoluble polymer called the “phosphotriester method”. The efficiency of this approach led to the first bioengineered genes; Somatostatin, human insulin, and human growth hormone were the first commercial products for a new startup company, Genentech.

In the 1970s, Professor Marvin Caruthers developed nucleoside phosphoramidite chemistry for the solid-phase synthesis of DNA and RNA, enabling precise and scalable production of oligonucleotides that remains foundational today. In parallel, biological research in the 1980s established how these molecules could modulate gene expression. Dr. Joseph Walder at Integrated DNA Technologies, Inc demonstrated that antisense oligonucleotides primarily inhibit gene expression by recruiting RNase H to cleave target RNA thereby linking sequence-specific hybridization to enzymatic RNA degradation (Walder and Walder, 1988) .

Then in the 1990’s the discovery of RNA interference by Dr. Andrew Fire and Dr. Craig Mello revealed a complementary pathway in which double-stranded RNA is processed into small interfering RNAs (siRNAs) that guide the RNA-induced silencing complex (RISC) to catalytically degrade target mRNA (Fire et al., 1998; Zamore, 2006). Within the framework of the central dogma, these insights converged into modern RNA-targeting therapeutic strategies. RNase H-mediated knockdown uses antisense oligonucleotides to direct enzymatic degradation of RNA, siRNA-based approaches leverage RISC for sequence-specific cleavage, and steric-blocking antisense oligonucleotides provide a versatile means of modulating RNA processing and function. Together with advances in chemical modification, these mechanisms underpin the versatility and programmability of contemporary oligonucleotide therapeutics.

Therapeutic Mechanisms

RNase H mediated



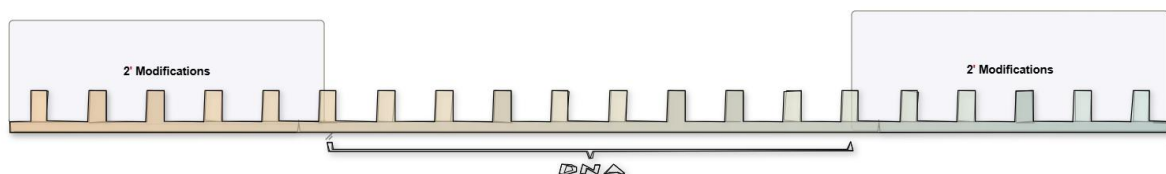
RNASE H ACTS AT THE TIME OF TRANSCRIPTION TO PREVENT R-LOOPS

RNase H is an endonuclease that plays a fundamental role in maintaining genome stability by selectively degrading the RNA strand of RNA–DNA hybrids. During DNA replication, it removes RNA primers required for initiation, ensuring their replacement with DNA and preserving genomic integrity. It also contributes to DNA repair by removing ribonucleotides, preventing structural instability and mutations (Eder, Walder and Walder, 1993). In addition, RNase H resolves R-loops that form during transcription, which, if accumulated, can interfere with replication and lead to DNA damage. Collectively, these functions establish RNase H as a key enzyme linking nucleic acid metabolism with the prevention of genome instability and disease (White, R.L. and Davis, R.W., 1976).



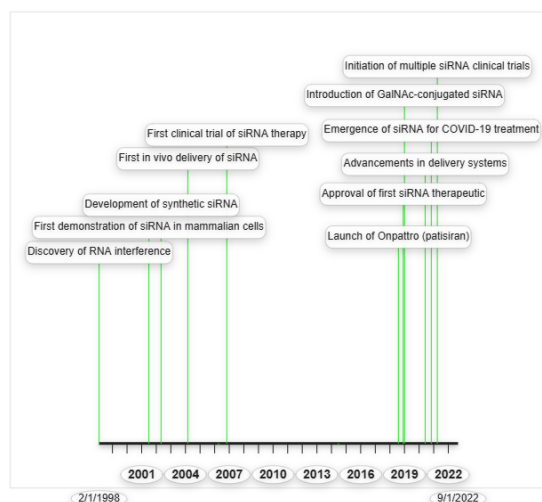
X-RAY DIFFRACTION HYBRID-BINDING DOMAIN OF HUMAN RNASE H1 IN COMPLEX WITH 12-MER RNA/DNA (PDB 3BSU) (NOWOTNY, 2008)

Antisense oligonucleotide (ASO) therapeutics use this biological mechanism for therapeutic purposes. A DNA molecule hybridizing to an RNA being transcribed forms the active substrate for RNase H and activates hydrolyses the phosphodiester backbone of the RNA strand. This catalytic reaction generates RNA fragments and prevents downstream gene products.



To harness this mechanism therapeutically, scientists needed to overcome the challenge of delivering DNA-like molecules to their targets without rapid degradation by nucleases. This led to the development of chemical modifications that preserve DNA properties of antisense oligonucleotides while enhancing their stability in biological systems. Early unmodified oligonucleotides were highly susceptible to nuclease degradation, severely limiting their effectiveness. This limitation was addressed through backbone modifications, most notably phosphorothioate linkages, which provide resistance to enzymatic degradation while retaining the ability to support RNase H activity. Additionally, sugar modifications such as 2'-O-methyl and 2'-O-methoxyethyl groups were introduced to improve binding affinity and further increase stability. These are often incorporated into “gapmer” designs, where a central DNA region capable of recruiting RNase H is flanked by modified nucleotides for protection. Collectively, these innovations enable antisense oligonucleotides to form stable RNA–DNA hybrids, activate RNase H, and achieve sufficient stability, bioavailability, and specificity for effective therapeutic application.

RISC mediated



RNAi FROM DISCOVERY TO THERAPY

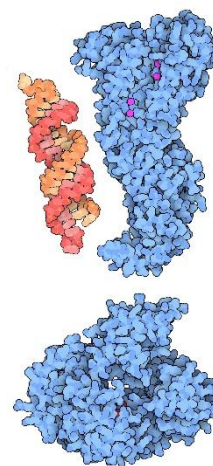


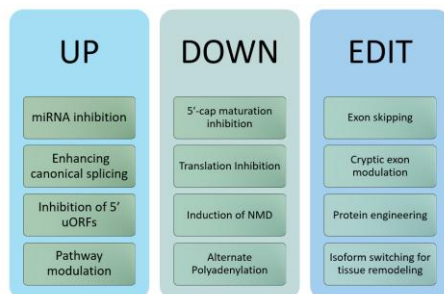
ILLUSTRATION BY DAVID S. GOODSSELL

Small interfering RNA (siRNA) is a powerful cellular defense system that helps protect organisms from viral infections and regulate gene expression. Cells are particularly sensitive to double-stranded RNA, which is often produced during viral replication and signals potential danger. In response, specialized proteins such as dicer (top right molecule in Goodsell's illustration) cut these double-stranded RNA into short fragments. These are then incorporated into argonaute proteins (bottom protein in Goodsell's illustration; PDB 248S), which use one strand as a guide to identify and destroy matching messenger RNA (mRNA) molecules. Through this process, known as RNA interference, cells can selectively silence specific genes or eliminate viral RNA (Fire, A., Xu, S., Montgomery, M.K., Kostas, S.A., Driver, S.E. and Mello, C.C. 1998).

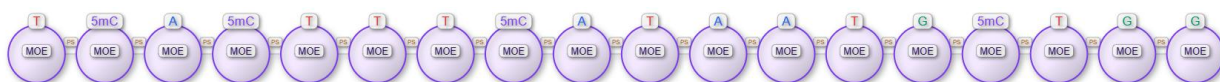
RNAi has become an important therapeutic strategy to selectively degrade target mRNA and inhibit protein expression. This approach enables precise targeting of disease-associated genes,

including those considered “undruggable” by conventional therapies, and has demonstrated long-lasting effects with infrequent dosing (Hu et al., 2020). Despite its promise, RNAi therapy faces challenges. These include off-target effects, immune activation, and efficient delivery to target tissues. Chemical modification and delivery systems, including lipid nanoparticles and molecular conjugates, have significantly improved the stability, specificity, and clinical applicability of siRNA-based drugs (Hu et al., 2020).

Steric-blocking



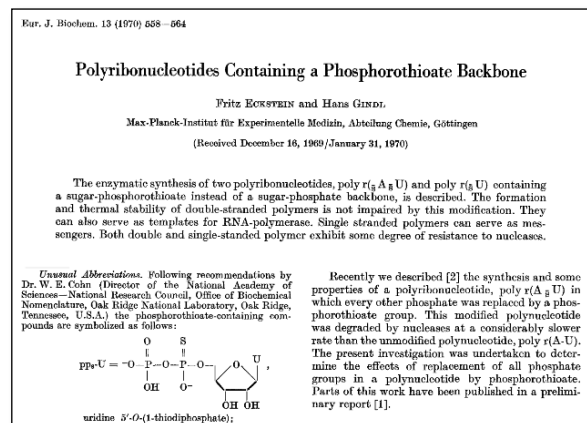
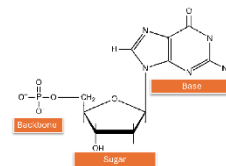
The steric-blocking strategy is the simplest mechanism of action but depending on the genomic context, it can offer a broad range of creative opportunities. The most famous steric blocking ASO is Nusinersen approved in December 2016 for spinal muscular atrophy.



SPINRAZA (nusinersen) is a chemically modified antisense oligonucleotide with a phosphorothioate backbone and 2'-O-methoxyethyl modifications that enhance stability of the SMN2 pseudogene. It is clinically indicated for the treatment of spinal muscular atrophy (SMA) in both pediatric and adult patients and is administered via intrathecal injection to ensure delivery to the central nervous system. The dosing regimen consists of four loading doses (three given at 14-day intervals followed by a fourth after 30 days), followed by a 12 mg maintenance dose every four months (Biogen, 2016).

Nusinersen acts as a splice-correcting antisense oligonucleotide that binds to the intronic splicing silencer ISS-N1 in SMN2 pre-mRNA, blocking repressor proteins and promoting exon 7 inclusion, thereby increasing production of functional SMN protein (Meijboom, Wood and McClorey, 2017). Clinically, this results in improved motor function and survival; however, treatment requires repeated lumbar punctures and monitoring due to potential adverse effects such as coagulation abnormalities, renal toxicity, and common events including respiratory infections and constipation (Biogen, 2016).

Backbone Chemistries



The importance of backbone chemistry in oligonucleotides dates to foundational work by [Eckstein and Gindl \(1970\)](#), who demonstrated that replacing a non-bridging oxygen in the phosphate backbone with sulfur (phosphorothioate) produced polyribonucleotide that retained the ability to form stable duplexes and serve as templates for RNA polymerase, while exhibiting enhanced resistance to nuclease degradation. This early work established the

principle that backbone modification could preserve biological function while improving stability.

Today, backbone engineering remains a central lever in oligonucleotide design, enabling control over stability, protein interactions, and pharmacokinetics. Phosphorothioate (PS) remains universal across ASO and many siRNA designs

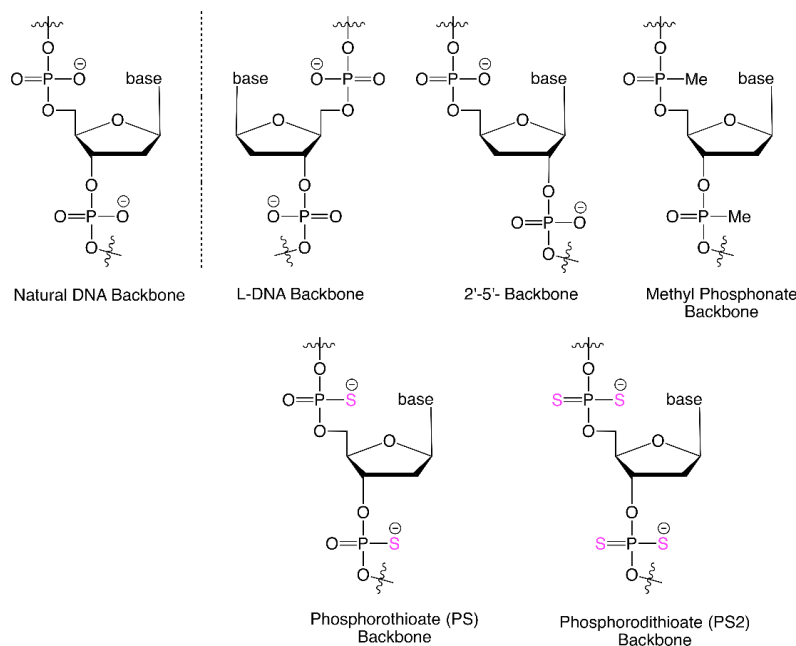
Charge, Stability and Function

Chemical modification of the oligonucleotide backbone underpins the success of modern nucleic acid therapeutics. While native phosphodiester (PO) linkages provide optimal compatibility with biological machinery, they are intrinsically unstable and pharmacologically limited. A diverse set of backbone and backbone-adjacent chemistries including phosphorothioate (PS), phosphorodithioate (PS₂), methylphosphonate (MP), phosphoramidate variants, phosphorodiamidate morpholino (PMO), and emerging phosphoramidate diester (PN) chemistries has emerged to address these constraints.

Backbone modifications operate within a constrained design space defined by three interdependent variables: charge, nuclease stability, and biological mechanism compatibility.

First, electrostatic charge governs interactions with proteins, membranes, and biological barriers. The canonical phosphodiester backbone is fully anionic, enabling strong protein binding and efficient recognition by endogenous enzymes, but simultaneously limiting membrane permeability and promoting nonspecific interactions. Phosphorothioate (PS) linkages retain this negative charge while altering its distribution, contributing to increased protein binding and improved pharmacokinetics. In contrast, several alternative chemistries including methylphosphonates, phosphoramidates, phosphorodiamidate morpholino (PMO), and PN linkages reduce or eliminate backbone charge. PMO and PN backbones are largely charge-neutral, fundamentally altering

electrostatic interactions, often reducing plasma protein binding and accelerating clearance while potentially improving intracellular distribution. However, reduced charge can also diminish hybridization affinity and weaken protein-mediated uptake pathways that are critical for productive delivery (Crooke, 2017; Juliano, 2016).



Second, nuclease stability is primarily governed by chemical substitutions at the phosphorus center and by steric protection from sugar modifications. Substitution of non-bridging oxygen atoms with sulfur (e.g., PS linkages) reduces nuclease-mediated hydrolysis and increases resistance to degradation in biological matrices (Lovric et al., 2025) (Wan et al., 2016).

Importantly, stability is not solely dependent on the presence of stabilizing chemistries but also on their distribution and combination. Mixed backbone designs (e.g., PO/PS or PS/PN chimeras) and gapmer architectures allow selective tuning of degradation kinetics and pharmacokinetics. For example, incorporation of PO linkages within a PS backbone reduces stability in a position-dependent manner, while sugar modifications such as MOE can compensate for reduced backbone protection and enable fine-tuning of tissue distribution (Lovrić et al., 2025; Geary, 2001).

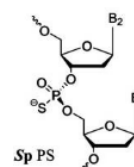
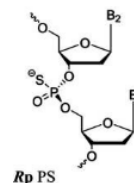
Emerging PN chemistries extend this paradigm by combining altered charge, stereochemical control, and compatibility with chimeric backbone designs. PN-containing oligonucleotides have demonstrated increased potency and durability in both in vitro and in vivo systems, suggesting that their benefits arise not only from nuclease resistance but also from improved cellular uptake and intracellular trafficking.

Taken together, these parameters define a tradeoff triangle in which no single chemistry simultaneously optimizes charge, stability, and function. Backbone chemistries such as PS, PMO, phosphoramidate, and PN, together with sugar modifications such as MOE and LNA, represent different strategies for navigating this space. Rather than replacing earlier chemistries, newer

approaches expand the combinatorial design landscape. Modern oligonucleotide therapeutics therefore rely on hybrid architectures—integrating multiple backbone and sugar modifications within a single molecule—to balance pharmacokinetics, potency, and mechanism.

Chirality

Substitution at the phosphorus center introduces chirality, generating Rp and Sp stereoisomers. In phosphorothioate backbones, each linkage can adopt either configuration, resulting in large mixtures of diastereomers in conventionally synthesized oligonucleotides. Chirality influences multiple aspects of oligonucleotide behavior. Rp linkages are often associated with improved RNase H recognition and RNA binding, whereas Sp linkages confer greater resistance to nuclease degradation. These opposing effects suggest that stereochemical control could, in principle, enable fine-tuning of backbone properties beyond what is achievable through chemistry alone.



Recent advances in stereocontrolled synthesis have enabled the generation of stereodefined oligonucleotides, allowing systematic exploration of these effects. This has led to the proposal of chirality as an additional design lever layered atop backbone chemistry.

Despite its conceptual appeal, experimental evidence indicates that stereochemical control provides limited practical benefits. Systematic studies demonstrate that while PS chirality modulates RNase H1 cleavage patterns in a predictable manner, these changes do not translate into improved potency or therapeutic index. Stereodefined oligonucleotides exhibit target knockdown comparable to stereorandom mixtures, with no configuration consistently outperforming the parent compounds. Indeed, fully Sp-modified backbones can exhibit substantial losses in activity.

At the mechanistic level, introduction of defined Rp and Sp linkages can bias cleavage site selection and, in some cases, produce dominant cleavage positions. Notably, such effects can be achieved with minimal stereochemical intervention, such as incorporation of a single RpSp step within the gap region. However, these alterations in enzymatic processing do not result in enhanced cellular or in vivo efficacy, underscoring a disconnect between biochemical precision and pharmacological outcome [REF Brad Wann].

Backbone modifications in the clinic

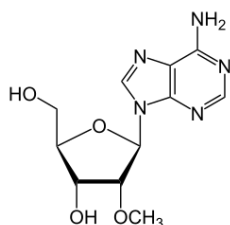
Chemistry	Pattern	Charge	Used by	In approved drugs?
PO	sooooooooooooooooo	full negative	Alnylam	Onpatro and others
PS	ssssssssssssssss	full negative	Ionis	Yes, many including Spinraza
PS + PO mix	soooooossssssssssooss	mixed	n-Lorem	Yes, neuro – gapmer
PN	sssnsssnsssnsssnss	reduced	Wave	
Stereopure PS		full negative	Wave	

Symbol Chemistry

o	phosphodiester (PO)
s	phosphorothioate (PS)
n	phosphoroamidate / PN

Furanose Sugar Modifications

2'-O-Methyl (2'-OMe)



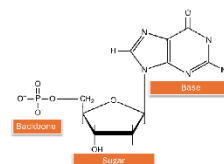
The “MOE” chemistry is one of the earliest and most widely adopted sugar modifications.

Benefits: - Improved nuclease resistance - Reduced immune activation - Moderate affinity enhancement.

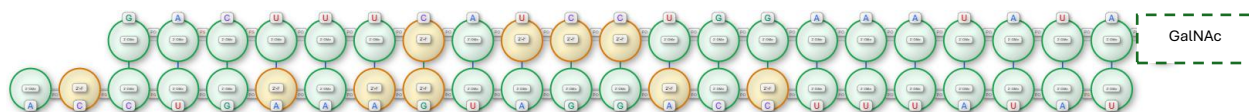
Common in both siRNA and single-stranded ASO platforms but more common in siRNA.

Endogenously, 2'-O-methylation is one of the most abundant and conserved RNA modifications, occurring across multiple RNA classes including tRNA, rRNA, mRNA, and small non-coding RNAs where it fine-tunes RNA structure, stability, and function. In tRNAs, 2'-OMe modifications are critical for accurate codon recognition. In rRNAs and mRNAs, 2'-OMe is found at the 5' cap and internally, where it regulates mRNA stability, translation efficiency, and discrimination between self and non-self RNA in innate immunity.

In therapeutics, this chemistry is central to siRNA design in Alnylam’s “enhanced stabilization chemistry” (ESC), where 2'-OMe combined with 2'-fluoro modifications improve pharmacokinetics,



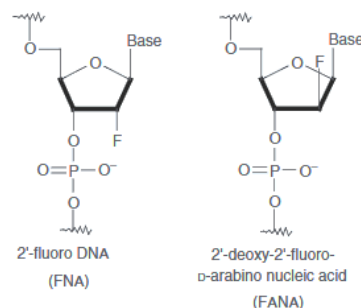
reduces off-target effects, and minimizes immune activation (Nair et al., 2014; Fitzgerald et al., 2017). These advances are clinically validated by approved drugs including Givosiran for acute hepatic porphyria, Lumasiran for primary hyperoxaluria type 1, and Inclisiran for hypercholesterolemia; all of which incorporate 2'-OMe-containing chemistries to enable durable gene silencing with infrequent dosing (Adams et al., 2020; Garrelfs et al., 2021; Raal et al., 2020).



Lumasiran (Oxlumo) is administered via subcutaneous and is an *ESC/GalNAc-conjugated* siRNA, it is designed for targeted delivery to hepatocytes following systemic administration, enabling RNA interference of the HAO1 gene in the liver. Efficacy was successfully achieved in the phase 3 ILLUMINATE-A trial, where Lumasiran caused a 65% mean reduction in 24-hour urinary oxalate compared to a 12% reduction in the placebo group; this is a statistically significant difference; 54% ($p < 0.0001$) (FDA, 2020). Approximately half of treated patients reached normalization of urinary oxalate levels by Month 6, indicating a clinically meaningful biochemical response (FDA, 2020). Consistent results were observed in younger patients in the ILLUMINATE-B study, where a 71% reduction in urinary oxalate (measured as oxalate-to-creatinine ratio) was reported, supporting efficacy across age groups (FDA, 2020). Across studies, reductions in urinary oxalate were rapid, occurring within the first few months of treatment and sustained over the study duration (FDA, 2020). The data demonstrates that Lumasiran provides substantial and consistent reductions in urinary oxalate (surrogate endpoint), supporting its efficacy in the treatment of primary hyperoxaluria type (FDA, 2020).

2'-Fluoro

2'-Fluoro modifications, including 2'-F-ANA (fluoro-arabino nucleic acid) and related FNA chemistries are highly potent nucleic acid modifications in which a fluorine atom replaces the 2'-hydroxyl group of the sugar. This change enhances nuclease stability and strengthens target binding, while also supporting steric-blocking mechanisms; in some cases, these modifications can enable gymnotic cellular uptake without delivery systems.



Benefits: - Strong target binding - Supports steric-blocking mechanisms - Enables gymnotic uptake in some cases. (Souleimanian et al., 2012)

	"Up" (Arabino configuration)	"Down" (Ribo configuration)
Sugar Pucker	Mostly O4'-endo (East) or C2'-endo	Strictly C3'-endo (North)
Helix Type	DNA-like (B-form)	RNA-like (A-form)
RNase H Activity	Activates RNase H	Does not activate RNase H
RNA Affinity	Higher than DNA; lower than 2'-F-RNA	Very high (extremely stable)



Inclisiran (Leqvio) is a chemically modified, double-stranded small interfering RNA (siRNA) conjugated to GalNAc to tissue-selectively knock down PCSK9 mRNA, leading to reduced hepatic PCSK9 synthesis and increased LDL receptor recycling, thereby lowering circulating low-density lipoprotein cholesterol (LDL-C) (FDA, 2021; EMA, 2020). It is indicated as an adjunct to diet and lipid-lowering therapy, including statins, for patients with primary hypercholesterolemia or atherosclerotic cardiovascular disease who require additional LDL-C reduction, and is administered subcutaneously with an extended dosing interval (initial dose, 3 months, then every 6 months) due to its long hepatic half-life (FDA, 2021; EMA, 2020). Nonclinical and clinical studies demonstrate dose-dependent and sustained reductions in PCSK9 and LDL-C, with distribution primarily to the liver, metabolism via nuclease degradation to shorter RNA fragments, and elimination mainly through urine and feces (FDA, 2021). Toxicological findings were largely non-adverse and associated with tissue accumulation, with no evidence of genotoxicity, carcinogenicity, or reproductive toxicity, supporting a favorable nonclinical safety profile (FDA, 2021)



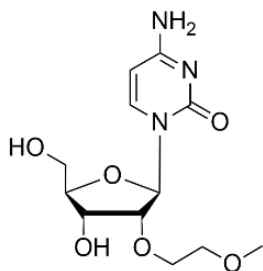
Givosiran (GIVLAARI) is a small interfering RNA (siRNA) therapeutic indicated for the treatment of adults with acute hepatic porphyria (AHP). It acts by targeting Aminolevulinate Synthase 1 (ALAS1) mRNA in hepatocytes, thereby reducing the accumulation of neurotoxic intermediates such as aminolaevulinic acid (ALA) and porphobilinogen (PBG). The drug is administered once a month through subcutaneous injection at a dose of 2.5 mg/kg (Alnylam Pharmaceuticals, Inc., 2019).

Clinical studies have demonstrated that Givosiran significantly reduces the frequency of porphyria attacks, with patients experiencing approximately 70% fewer attacks compared to placebo in the ENVISION trial (*page 10 on the drug label/US Prescribing Information (USPI)*). Common adverse effects

include nausea and injection-site reactions, while more serious risks include hepatic toxicity, renal impairment, and rare anaphylactic reactions (Alnylam Pharmaceuticals, Inc., 2019).

2'-F-ANA (2'-fluoro-arabino nucleic acid) is a chemically modified nucleotide in which substitution at the 2' position of the sugar enhances nuclease resistance and increases binding affinity to complementary RNA, while maintaining compatibility with RNase H activity in gapmer designs (Souleimanian et al., 2012). These properties have made 2'-F-ANA a useful modification in preclinical studies, where it has demonstrated potent and sustained gene silencing. However, despite its favorable biochemical profile, this chemistry has not yet been translated into clinically approved oligonucleotide therapeutics and remains primarily confined to experimental applications.

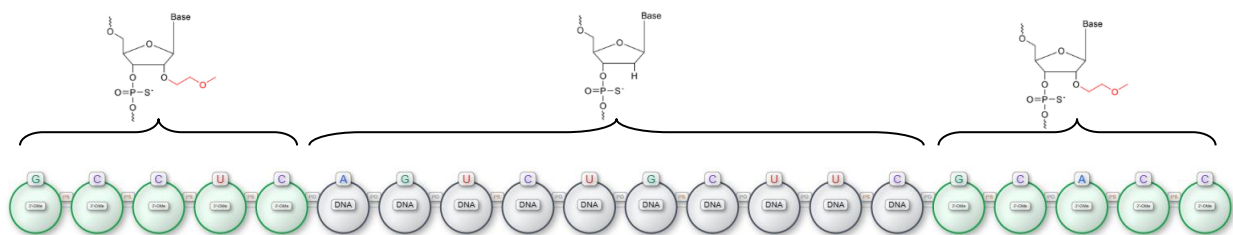
2'-O-Methoxyethyl (2'-MOE) / MOE chemistry



Benefits: - High binding affinity - Strong nuclease resistance - Favorable safety profile – long duration of action

The development of 2'-O-methoxyethyl (MOE) chemistry as a therapeutic originated from limitations in antisense oligonucleotide (ASO) technology; unmodified nucleic acids were rapidly degraded and exhibited poor pharmacokinetic properties. In 1995, Pierre Martin, a chemist working at Ciba-Geigy (now Novartis) in Basel, Switzerland, reported the synthesis of 2'-O-alkylated ribonucleosides, including the 2'-O-(2-methoxyethyl) (MOE)

modification, in a publication in *Helvetica Chimica Acta*: Martin P. A New Access to 2'-O-Alkylated Ribonucleosides and Properties of 2'-O-Alkylated Oligoribonucleotides. Helvetica Chimica Acta. 1995 Mar;78(2):486-504. DOI: 10.1002/hlca.19950780219. (Hill & Hall, 2023). As antisense technology evolved, MOE modification became a hallmark of second-generation ASOs, building upon phosphorothioate backbone modifications to further enhance potency, tissue residence time, and tolerability (Zanardi et al., 2018). Over time, these combined chemical innovations enabled the successful clinical translation of antisense therapeutics, with MOE-modified oligonucleotides now forming the basis of several approved drugs and numerous candidates in development.



The MOE Modification of RNA Gapmer: Mipomersen

Inotersen is administered as a once-weekly subcutaneous injection and has demonstrated clinical efficacy in phase II/III NEURO-TTR randomized controlled trial. The study showed that inotersen

significantly slowed the progression of neuropathy and improved quality of life compared with placebo, as measured by the modified Neuropathy Impairment Score and the Norfolk Quality of Life–Diabetic Neuropathy questionnaire (CADTH, 2020). However, despite achieving statistical significance, the clinical meaningfulness of these improvements remains uncertain due to the absence of clearly defined thresholds for clinically important differences (CADTH, 2020).



Safety remains a key concern, as treatment is associated with a high incidence of adverse events, including thrombocytopenia, renal impairment, and injection-site reactions. Serious adverse events and deaths have been reported in clinical trials, necessitating regular monitoring of platelet counts and renal function (CADTH, 2020). Clinical guidance further emphasizes the importance of ongoing monitoring through regular blood and urine tests to assess both efficacy and toxicity (Mayo Clinic, n.d.).

Inotersen can reduce platelet counts, increasing the risk of bleeding and infection, and patients are advised to take precautions to minimize injury and exposure to infectious agents (Mayo Clinic, n.d.). It has also been associated with serious organ-related complications, including kidney and liver toxicity, with symptoms such as reduced urine output, swelling, jaundice, or abdominal pain requiring immediate medical attention (Mayo Clinic, n.d.). In addition, there is an elevated risk of inflammatory or immune-mediated conditions, stroke, and arterial dissection, highlighting the need for vigilance regarding neurological symptoms (Mayo Clinic, n.d.).

Hypersensitivity reactions, including life-threatening allergic responses, have also been reported, alongside reductions in vitamin A levels that may lead to visual disturbances, necessitating ophthalmologic monitoring in some patients (Mayo Clinic, n.d.). Patients are further advised to consult their healthcare provider before taking any concomitant medications, including over-the-counter or herbal products, due to potential interactions (Mayo Clinic, n.d.).

Overall, Inotersen represents an important advancement in RNA-targeting therapeutics and offers a more convenient, self-administered alternative to intravenous treatments. However, its use is limited by safety concerns and ongoing uncertainties regarding long-term efficacy and broader clinical benefits (CADTH, 2020; Rüger et al., 2020).

MOE-modified ASOs are widely used in two RNA-targeting strategies: RNase H-mediated RNA degradation and splice modulation via steric blocking. These two approaches enable treatment of a broad range of genetic diseases (Sang et al., 2024). For example, MOE-containing gapmer ASOs such as Inotersen and Eplontersen are used in the treatment of hereditary transthyretin amyloidosis (hATTR), where they bind transthyretin (TTR) mRNA and promote its degradation, thereby reducing pathogenic protein accumulation (Sang et al., 2024). Similarly, Tofersen, a 5'-10'-5' MOE-modified gapmer ASO approved for amyotrophic lateral sclerosis (ALS), targets mutant SOD1 mRNA, demonstrating the utility of MOE chemistry in neurodegenerative disease through selective reduction of toxic protein expression (Sang et al., 2024). In the context of splice modulation, Nusinersen (Spinraza) discovered at Cold Spring Harbor uses a fully MOE-modified ASO to promoting inclusion of exon 7 in SMN2 mRNA and thereby restoring production of functional SMN protein (Sang et al., 2024). Beyond these established therapies,

MOE chemistry has also enabled highly personalized approaches, exemplified by Milasen, a patient-specific ASO designed to correct aberrant splicing in Batten disease, highlighting the adaptability of MOE-modified oligonucleotides for precision medicine (Sang et al., 2024). This chemistry is heavily used by nLore for their nano-rare therapeutics programs.



5-10-5 MOE GAPMER WITH ESC siRNA OVERLAP IN BOTH APPROVED FOR ATTR AMYLOIDOSIS. THESE FDA APPROVED DRUGS OVERLAP IN SEQUENCE BUT CONTAIN VERY DIFFERENT CHEMISTRIES AND DIFFERENT MECHANISM OF ACTION

MOE-modified ASOs are widely used in two principal RNA-targeting strategies: RNase H-mediated RNA degradation and splice modulation via steric blocking, enabling treatment of a broad range of genetic diseases (Sang et al., 2024). For example, MOE-containing gapmer ASOs such as Inotersen and Eplontersen are used in hereditary transthyretin amyloidosis (hATTR), where they bind transthyretin (TTR) mRNA and promote its degradation (Sang et al., 2024). Similarly, Tofersen, a MOE-modified gapmer ASO approved for amyotrophic lateral sclerosis (ALS), targets mutant SOD1 mRNA to reduce toxic protein expression (Sang et al., 2024). In the context of splice modulation, Nusinersen (Spinraza) employs a fully MOE-modified ASO to promote inclusion of exon 7 in SMN2 mRNA, restoring functional SMN protein (Sang et al., 2024). Beyond these established therapies, MOE chemistry has enabled highly personalized approaches such as Milasen, a patient-specific ASO designed to correct aberrant splicing in Batten disease (Sang et al., 2024). The continued evolution of MOE chemistry, particularly when combined with advanced delivery strategies and conjugation technologies, underscores its central role in enabling durable, tissue-targeted, and precision RNA therapeutics.

Fomivirsen (Vitraven®)	Phosphorothioate backbone (PS)	Cytomegalovirus (CMV) retinitis	1999
Mipomersen (Kynamro®)	PS & 2'-MOE	Homozygous familial hypercholesterolemia	2013
Nusinersen (Spinraza®)	PS & 2'-MOE	Spinal muscular atrophy	2016
Inotersen (Tegsedi®)	PS & 2'-MOE	Hereditary transthyretin-mediated amyloidosis	2018
Milasen	PS & 2'-MOE	CLN7 gene mutation associated with Batten disease	2018
Volanesorsen (Waylivra®)	2'-MOE	Familial chylomicronemia syndrome	EMA approved 2019
Tofersen (Qalsody®)	PS & 2'-MOE	Amyotrophic lateral sclerosis with SOD1 mutation	2023
Eplontersen (Wainua®)	PS, 2'-MOE & GalNAc	Polyneuropathy of hereditary transthyretin-mediated amyloidosis	2023

Incorporation of MOE at the 2'-position of the ribose increases conformational rigidity and enhances hybridization while reducing susceptibility to nuclease-mediated cleavage (Zanardi et al., 2018). From a pharmacokinetic and pharmacodynamic (PK/PD) perspective, MOE-modified ASOs exhibit slow plasma clearance, high plasma protein binding, and extensive tissue distribution—particularly to the liver, kidney, and adipose tissue—resulting in long tissue half-lives that can extend from weeks to months (Takakusa et al., 2023). For example, 2'-MOE ASOs such as Mipomersen demonstrate terminal half-lives of ~30 days in plasma and similar persistence in liver tissue, reflecting equilibrium between plasma and tissue compartments and supporting infrequent dosing regimens (Takakusa et al., 2023). These prolonged kinetics arise from strong protein binding and intracellular retention following endocytosis and endosomal trafficking, which enable sustained pharmacological activity even after plasma levels decline (Takakusa et al., 2023).

Importantly, pharmacodynamic effects are more closely correlated with tissue concentrations than plasma levels, highlighting the importance of tissue PK in dose optimization and therapeutic efficacy (Christensen et al., 2025). Furthermore, advances in chemical stabilization and conjugation strategies have enabled improved delivery beyond the liver, expanding the therapeutic reach of oligonucleotides to extrahepatic tissues such as muscle, brain, and lung (Anand et al., 2025).

Volanesorsen (Waylivra): ApoC-III targeting ASO for familial chylomicronemia syndrome (FCS)



Base: A G mC mU mU mC T T G T mC mC A G mC mU mU mU A mU

Backbone Full phosphorothioate (PS)

Sugar: E E E E E d d d d d d d d E E E E E

Sugar legend

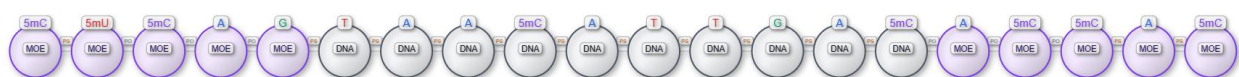
E = 2'-MOE

d = DNA (2'-deoxy)

Volanesorsen demonstrated substantial triglyceride-lowering efficacy but is associated with several safety considerations. The most clinically significant risk is thrombocytopenia, including cases of severe platelet reduction. In addition, injection-site reactions are very common, occurring in a substantial proportion of administrations, although these are generally mild to moderate and primarily affect tolerability rather than safety. Rare but serious immune-mediated events, such as serum sickness, have also been reported, indicating potential for hypersensitivity reactions. (Witztum et al., 2019; Tardif et al., 2016).

Olezarsen, a GalNAc-conjugated antisense oligonucleotide version of Volanesorsen and was developed to solve its relatively poor safety profile. Thrombocytopenia with Volanesorsen was attributed to high systemic exposure and dosing burden. By incorporating N-acetylgalactosamine (GalNAc) for hepatocyte-targeted delivery, Olezarsen matches efficacy with less frequent doses, thereby reducing systemic exposure. Clinical data demonstrate that while some adverse events persist, including mild thrombocytopenia and liver enzyme elevations, these effects are dose-dependent and more pronounced at higher doses, with overall safety outcomes approaching those of placebo. This supports the concept that dose and exposure are central determinants of antisense oligonucleotide toxicity, and that GalNAc conjugation can effectively improve the therapeutic index by decoupling efficacy from systemic toxicity (Marston et al., 2026; Witztum et al., 2019).

	Volanesorsen	Olezarsen (GalNAc)
Dosing	High dose (e.g., 300 mg weekly → biweekly)	Lower dose (50–80 mg monthly)
Delivery	Systemic (no targeting)	Hepatocyte-targeted (GalNAc)
Thrombocytopenia	Clinically significant; severe cases reported	Less frequent; dose-dependent, higher at 80 mg
Injection-site reactions	Very common (~24% of injections)	Less frequent (~10–16%)
Liver enzyme elevations	Not dominant signal	Increased, especially at higher dose (80 mg)
Hepatic fat	No consistent increase	Dose-dependent increase observed
Hypersensitivity / immune	Rare (e.g., serum sickness)	Mild hypersensitivity events slightly increased
AE rate	Higher discontinuation vs placebo	Similar to placebo overall
Key safety driver	Systemic exposure / high dose	Residual dose-dependent toxicity, reduced by targeting



IONIS-HTTRx (Tominersen, Roche/Ionis) – MOE-modified “gapmer” ASO targeting huntingtin mRNA for Huntington’s disease.

Base **m5C m5U m5C A G T A A m5C A T T G A m5C A m5C m5C A m5C**

Backbone **PS – PO – PO – PO – PS – PS – PS – PS – PS – PS – PS – PS – PS – PS – PS – PS – PO – PO – PO – PS**

Sugar **E E E E E d d d d d d d d E E E E E → 5 – 10 – 5 MOE gapmer**

Legend **E = 2'-MOE; d = DNA**

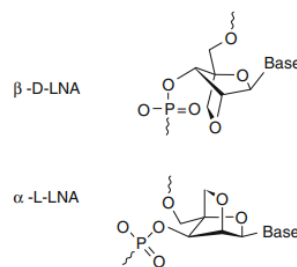
In the Phase III GENERATION HD1 trial, patients received Tominersen via intrathecal injection at doses of 120 mg every 8 weeks or every 16 weeks. Although the drug successfully lowered huntingtin protein levels, it failed to improve clinical outcomes and in some cases worsened disease progression, particularly with more frequent dosing. This lack of efficacy is thought to result from non-selective reduction of both mutant and normal huntingtin along with possible overdosing and intervention occurring too late in the disease course.

Locked Nucleic Acids (LNA)

Benefits: - Strong target binding but potentially narrow therapeutic window.

Locked nucleic acids were introduced in the late 1990s:

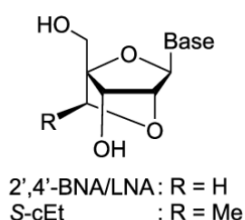
Obika, Satoshi; Nanbu, Daishu; Hari, Yoshiyuki; Morio, Ken-ichiro; In, Yasuko; Ishida, Toshimasa; Imanishi, Takeshi (1997-12-15). *"Synthesis of 2'-O,4'-C-methylenuridine and -cytidine. Novel bicyclic nucleosides having a fixed C3', -endo sugar pucker"*. *Tetrahedron Letters*. 38 (50): 8735–8738. Doi:10.1016/S0040-4039(97)10322-7. ISSN 0040-4039.



These nucleosides incorporate a 2'-O,4'-C methylene bridge that locks the ribose ring into a C3'-endo conformation characteristic of RNA duplexes (Koshkin et al., 1998; Obika et al., 1998).

This conformational constraint produces several key effects:

- Strongly enhanced hybridization affinity
- Increased thermal stability of duplexes
- Improved mismatch discrimination
- Terminal LNA incorporation increases stability approximately tenfold over unmodified
- Resist exonuclease degradation when flanking residues are modified (Kurreck et al., 2002)



Unlike earlier modifications, LNA technology was a shift toward structure-driven nucleic acid engineering, where conformational preorganization directly enhances function.

Constrained ethyl (cEt) chemistry is a type of LNA modification that incorporates a 2'-O-ethyl substituent within a constrained bicyclic framework; used primarily by Ionis Pharmaceuticals. Structural analyses show that cEt and LNA adopt highly similar sugar conformations, while cEt positions its substituents closer to the phosphate backbone, contributing to enhanced resistance to enzymatic degradation (Pallan et al., 2012).

	ΔT_m per modification (°C)	Relative affinity vs DNA
DNA (unmodified)	Baseline	1×
Phosphorothioate (PS DNA)	0	0.8–1×
2'-O-methyl (2'-OMe)	+0.5 to +1.0	1.2–1.5×
2'-fluoro (2'-F)	+1.0 to +2.0	1.5–2×
2'-O-methoxyethyl (2'-MOE)	+0.9 to +1.7	1.5–2×
Locked nucleic acid (LNA)	+3.5 to +8.0	3–10×

LNA as Bio-Tools

Locked nucleic acid (LNA) chemistry has become an important component of modern PCR-based technologies, particularly in applications requiring high specificity, sensitivity, and reproducibility. The enhanced hybridization affinity conferred by LNA modifications enables the design of short oligonucleotide primers and probes with elevated and tunable melting temperatures (T_m), which improves assay performance across a wide range of targets (Braasch and Corey, 2001; Grunweller and Hartmann, 2007).

Incorporation of LNA residues into PCR primers and fluorescent probes:

- Improved binding to difficult or GC-rich templates
- Shorter primer design without loss of affinity
- Enhanced mismatch discrimination
- Enhanced fluorescence signal upon hybridization
- Enable stable hybridization to short targets

Strategic placement of LNA nucleotides often near the 3'-end can significantly improve allele-specific amplification, making LNA-modified primers particularly useful for single nucleotide polymorphism (SNP) detection and genotyping assays (Di Giusto and King, 2004; Mouritzen et al., 2003). The increased thermodynamic stability also reduces nonspecific amplification by favoring correct primer–template pairing. (Morandi et al., 2007; Valoczi et al., 2004; Castoldi et al., 2006).

LNA in Therapeutics

Miravirsen is a first-in-class LNA-modified antisense oligonucleotide that targets miR-122, a liver microRNA essential for HCV replication, leading to durable, dose-dependent viral suppression with

a strong safety profile and no observed resistance. Its success validated LNA chemistry as a viable and safe platform for RNA-targeting therapeutics.

Miravirsen (Santaris/Roche) – anti-miR-122 (HCV)

L d L d d L L d d L d L d L L → L = LNA ; d = DNA

Miravirsen is a 15-nucleotide LNA-modified phosphorothioate antisense oligonucleotide designed to bind and sequester miR-122, a liver-specific microRNA essential for hepatitis C virus (HCV) replication. By forming a highly stable heteroduplex with miR-122, miravirsen functionally inhibits the microRNA, thereby destabilizing HCV RNA and suppressing viral replication.

miR-122 binds to two conserved sites in the 5' untranslated region of the HCV genome, enhancing viral stability and replication. Miravirsen blocks this interaction.

Clinical Data (Phase 2a):

- Dosing: Five weekly subcutaneous injections (3, 5, or 7 mg/kg)
- Demonstrated dose-dependent reductions in HCV RNA:
- 1.2 log₁₀ IU/mL reduction (3 mg/kg)
- 2.9–3.0 log₁₀ IU/mL reduction (5–7 mg/kg)
- Viral suppression persisted beyond the dosing period
- Undetectable HCV RNA observed in multiple patients at higher doses

Key Differentiators:

- No evidence of viral resistance or escape mutations
- Activity across HCV genotypes due to conserved miR-122 binding sites
- Favorable pharmacokinetics with prolonged tissue half-life
- Lack of cytochrome P450 interactions, reducing drug–drug interaction risk

Safety Profile:

- No dose-limiting toxicities observed
- Mostly mild to moderate adverse events (e.g., headache, fatigue)
- No significant hepatotoxicity or clinically relevant laboratory abnormalities
- Reduction in cholesterol levels consistent with miR-122 biology

Miravirsen was a first-in-class microRNA-targeting therapeutic that showed benefit and validated the LNA chemistry as a safe clinical construct for RNA-targeting drugs.

Cobomarsen is an LNA-based anti-miR-155 therapy for cutaneous T-cell lymphoma that showed encouraging early clinical results, including a favorable safety profile and disease improvement in most patients, supporting its potential as a novel microRNA-targeted cancer treatment.

L d L d d L L d L d L L L → L = LNA ; d = DNA (phosphonothioate backbone)

Phase 1 clinical trial evaluated Cobomarsen in patients with mycosis fungoides (MF), the most common subtype of CTCL, assessed safety, tolerability, pharmacokinetics, and preliminary efficacy across multiple routes of administration, including intralesional, subcutaneous (SC), and intravenous (IV) delivery at doses up to 900 mg.

Cobomarsen demonstrated a favorable safety profile, with no serious adverse events attributed to the drug. The most commonly reported AE was fatigue, neutropenia, injection site pain, nausea, pruritus, and headache. Dose-limiting toxicities were limited to Grade 3 pruritus and tumor flare, and the maximum tolerated dose was not reached.

91% showed improvement in disease burden as measured by the modified Severity Weighted Assessment Tool mSWAT -- the percentage of Body Surface Area affected by patches, plaques, and tumors.

Importantly, improvements in clinical endpoints were accompanied by enhanced quality of life and meaningful patient benefits beyond tumor reduction. These results highlight the therapeutic potential of LNA-based anti-miR strategies in oncology.

[TODO]



RG-101 is a GalNAc-conjugated anti-miR-122 therapy that showed strong antiviral activity and good initial tolerability but later revealed safety concerns due to transient hyperbilirubinemia from off-target effects. This led to the development of RG6650, an optimized follow-up compound designed to maintain efficacy while improving safety by reducing interference with bilirubin transport.

RG-101 LNA/MOE/DNA Clinically strong antiviral, but safety issue

RG6650 cEt/DNA selected for improved safety

In early clinical studies (phase 1B study), RG-101 was well tolerated after single-dose administration, with no serious adverse events and mostly mild, self-limiting side effects such as fatigue and injection site reactions (van der Ree et al., 2017). Transient increases in alkaline phosphatase and a single case of cholestasis were observed, but there was no consistent evidence of hepatocellular injury or significant bilirubin elevation across the cohort (van der Ree et al., 2017).

Later studies of RG-101 demonstrated strong antiviral efficacy but showed transient hyperbilirubinemia that was attributed to off-target inhibition of the MRP2 transporter (van der Ree et al., 2017; Drygin, 2022). Following this finding, Regulus conducted a focused optimization effort to identify next-generation compounds that retained potent miR-122 inhibition while minimizing interference with bilirubin transport (Drygin, 2022). This work involved a screening and structure–activity relationship (SAR) campaign that ultimately leading to the identification of RG6650 (Drygin, 2022).

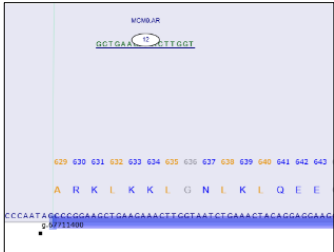
Feature	RG-101	RG6650 / RG7443
Base	A m5C A m5C m5C A m5U T G U C A C A C T C C A	U C A C A C T C C
Backbone	Full phosphorothioate	Full phosphorothioate
Sugar	E E E E E E E d d L L d L d L d L L d	K K d K d K d K K
Legend	E = 2'-MOE; d = DNA; L = LNA; A = adenine; T = thymine; G = guanine; C = cytosine; U = uracil; m ⁵ C = 5-methylcytosine; m ⁵ U = 5-methyluracil.	K = S-cEt; d = DNA
Conjugation	3' GalNAc-conjugated	3' GalNAc-conjugated

Both EZN-4176 (Enzon) and ISIS-AR Rx/AZD5312 (Ionis) were designed to suppress androgen receptor (AR) expression in castration-resistant prostate cancer, but they differ in chemistry and failure profile. EZN-4176 is a first-generation LNA gapmer with very high binding affinity, which likely contributed to hybridization-dependent off-target effects, including binding to “unintended transcripts” such as the DNA repair gene MCM9 (see figure below), alongside the expected AR target. This broad binding profile, combined with the inherent potency of LNA chemistry, is consistent with the hepatotoxicity and limited clinical activity observed in the phase I study.



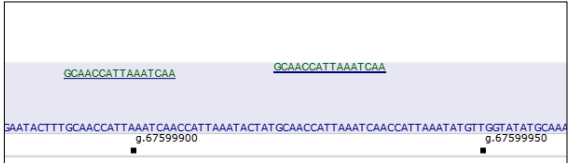
In contrast, **ISIS-AR Rx (AZD5312)** uses a later cEt-modified gapmer design intended to retain potency while improving tolerability and specificity; however, despite better chemical optimization, it also failed to demonstrate sufficient efficacy in clinical development. Together, these compounds illustrate a key challenge in antisense therapeutics: increasing binding affinity (as with LNA) can enhance potency but may also increase off-target interactions and toxicity, while newer chemistries like cEt improve safety margins yet do not guarantee clinical success if target engagement or biological effect remains insufficient.

	EZN-4176	ISIS-AR Rx (AZD5312)
Developer	Enzon	Ionis / AstraZeneca
Chemistry	LNA gapmer	cEt gapmer
Target	Androgen receptor (AR)	Androgen receptor (AR)
Backbone	Phosphorothioate	Phosphorothioate
Sugar mods	LNA + DNA	cEt + DNA
Clinical outcome	ALT elevations (toxicity signal)	Developed to improve potency/safety



The Enzon–Santaris LNA gapmer program progressed into the clinic by targeting oncogenic drivers like AR and PI3K/Akt, but the same compound may have also unintentionally suppressed MCM9, a DNA repair gene linked to cancer predisposition. --a critical failure in bioinformatics off-target analysis that made it all the way into the clinic!

Feature	ISIS-AR Rx / AZD5312 (AKA ISIS 560131)
Base	TTGATTTAATGGTTGC
Backbone	Full phosphorothioate
Sugar	k k k d d d d d d d d k k E
Legend	k = (S)-cEt; d = DNA; E = 2'-MOE; C = 5-methylcytosine



ISIS AR Rx HITS ANDROGEN RECEPTOR TWICE.

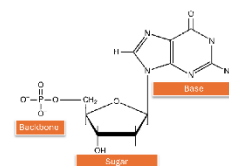
Recent data (2024–early 2026) indicate that AZD5312 shows promising efficacy in prostate cancer with observed PSA declines in some heavily pre-treated patients and enhanced activity in combination with AKT inhibitors (e.g., Capivasertib). Its safety profile is consistent with antisense

oligonucleotides, with common adverse events including fatigue, injection site reactions, and flu-like symptoms, while more serious but manageable risks include thrombocytopenia, hepatotoxicity, and renal effects requiring routine monitoring (AstraZeneca, 2024; ClinicalTrials.gov, 2025; de Bono et al., 2023).

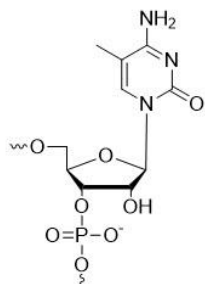
Locked nucleic acid (LNA) chemistry represents one of the most potent antisense modification strategies for RNA targeting due to its conformationally “locked” sugar, which markedly increases binding affinity and duplex stability relative to earlier chemistries such as 2'-O-methoxyethyl (MOE) and 2'-O-methyl modifications (Qassem et al., 2024; Crooke et al., 2018). This enhanced affinity enables shorter oligonucleotides, improved potency, and flexible designs (e.g., LNA/DNA gapmers) that retain RNase H activity while achieving stronger target engagement than MOE-based ASOs (Qassem et al., 2024). However, these advantages come with important trade-offs: compared with MOE chemistries, which generally exhibit lower toxicity and reduced immunogenicity, LNA-modified ASOs show a higher incidence of hybridization-dependent off-target effects due to their increased binding strength and shorter sequence length, as well as a greater risk of hepatotoxicity observed in preclinical models (Qassem et al., 2024; Crooke et al., 2018). Thus, while MOE modifications offer a more balanced safety profile with moderate potency, LNA chemistry shifts the balance toward maximal efficacy at the expense of a narrower therapeutic window, making careful sequence design and dose optimization critical for clinical application.

Nucleobase modifications

“Base” chemistry is increasingly used to modulate recognition, stability, and enzymatic interactions. While there are numerous one-off chemistries this review will cover some of the more commonly used modifications.



5-methylcytidine (m5C): Improves stability and reduces immunogenicity

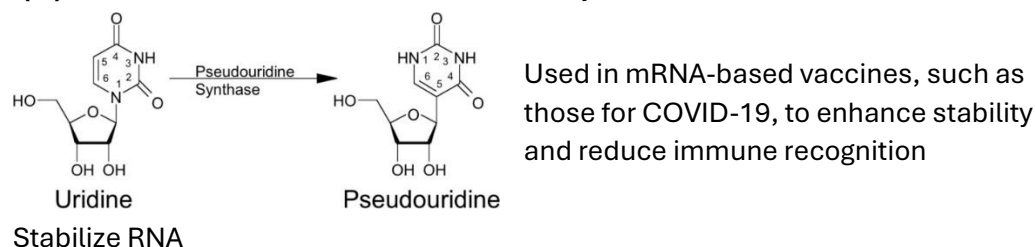


5-methylcytosine (m5C; 5-methylcytidine, 5-Me-C) is a methylated RNA base analog found across tRNA, other cellular RNAs, where it functions as a post-transcriptional modification contributing to RNA stability, processing, and translation (Motorin and Helm, 2011; Squires et al., 2012). Its best-characterized role is in tRNA, where m5C protects transcripts from endonucleolytic cleavage and preserves structural integrity under stress (Blanco et al., 2014).

Incorporation of 5-methylcytidine into synthetic RNA oligonucleotides enhances base pairing and duplex stability through increased hydrophobicity at the C5 position and raises melting temperature (~1–1.5 °C per modification). This stabilizing effect improves hybridization, supports disruption of target secondary structure, and enables the use of shorter antisense oligonucleotides with improved binding efficiency. At the same time m5C reduces susceptibility to nuclease

degradation and dampens activation of innate immune sensors that preferentially recognize unmodified RNA (Karikó et al., 2005; Anderson et al., 2010).

Pseudouridine (Ψ): enhances RNA-like behavior and stability

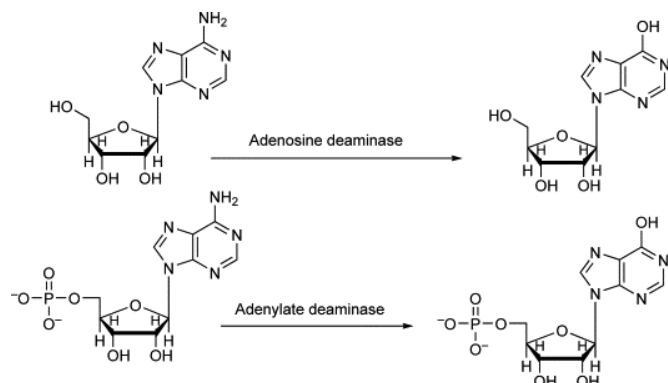


Pseudouridine (Ψ), the C5-glycoside isomer of uridine, is the most abundant post-transcriptional RNA modification and is widely distributed across tRNA, rRNA, mRNA, and non-coding RNAs, where it contributes to RNA stability, structure, and translation (Lin et al., 2021). The isomerization of uridine to Ψ introduces a carbon–carbon glycosidic bond and an additional hydrogen bond donor, enhancing base stacking, strengthening RNA–RNA and RNA–protein interactions, and increasing structural rigidity (Lin et al., 2021). Endogenously, Ψ plays critical roles in stabilizing tRNA architecture, supporting ribosome function through rRNA modification, and modulating mRNA stability and translation efficiency (Zhao and He, 2015; Lin et al., 2021). These native properties drive their use in RNA therapeutics, where incorporation of Ψ into synthetic RNAs enhances duplex stability, increases resistance to nuclease degradation, and improves translational output, often yielding significantly higher protein expression compared to unmodified RNA (Lin et al., 2021).

Pseudouridine also reduces immunogenicity by diminishing activation of innate immune sensors and downstream cytokine responses, thereby enabling sustained translation and improved tolerability of exogenous RNA (Karikó et al., 2008; Lin et al., 2021). This combination of structural stabilization, enhanced translation, and immune evasion has made pseudouridine and its derivatives (e.g., N1-methylpseudouridine) foundational to modern RNA therapeutic platforms, including mRNA vaccines and protein replacement therapies, where mimicking endogenous RNA chemistry is key to achieving clinical efficacy.

mRNA COVID-19 vaccines are structurally composed of a nucleoside-modified mRNA encoding the SARS-CoV-2 spike protein, encapsulated within a lipid nanoparticle (LNP) delivery system. The mRNA includes regulatory elements such as 5' and 3' untranslated regions, a coding sequence for a stabilised pre-fusion spike protein, and multiple stop codons to ensure accurate translation (Morais, Adachi and Yu, 2021). A key structural feature is the substitution of uridine with N1-methyl-pseudouridine, which enhances mRNA stability, reduces innate immune activation, and improves protein translation efficiency (Morais, Adachi and Yu, 2021). Surrounding this, the LNP structure protects the mRNA from degradation, facilitates cellular uptake, and promotes delivery to immune cells such as dendritic cells, enabling antigen expression and subsequent immune activation (Morais, Adachi and Yu, 2021)

Inosine: enables flexible base pairing, critical for editing systems; Inosine is used in ADAR technology (RNA editing) since it is read by cells as a G. Inosine is a functionally unique RNA



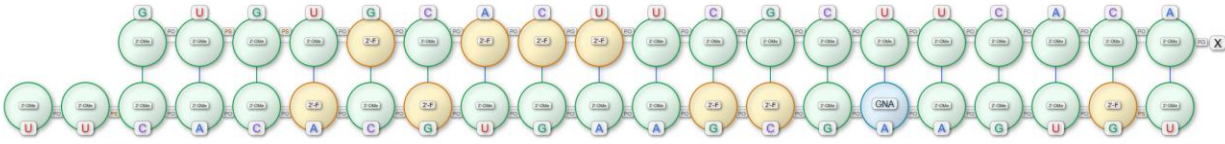
modification generated through ADAR-mediated deamination of adenosine, enabling dynamic rewriting of RNA

information without altering the genome. Because inosine is interpreted as guanosine during translation and splicing, A-to-I editing can recode proteins, alter splice sites, and reshape RNA structure, thereby modulating transcript stability, localization, and protein output (Srinivasan et al., 2021 ; Eisenberg and Levanon, 2018). In tRNAs, inosine at the

wobble position expands codon recognition capacity, allowing a single tRNA to decode multiple synonymous codons and enhancing translational efficiency and proteome diversity (Srinivasan et al., 2021). In mRNAs, site-specific editing can produce amino acid substitutions or regulate RNA processing, as exemplified by editing of neurotransmitter receptors and ion channels, with direct implications for neurological function and disease.

This intrinsic property provides the platform for RNA editing therapeutics; programmable A-to-I editing uses natural ADAR enzymes to fix disease-causing mutations at the RNA level, making it a temporary and potentially safer alternative to editing DNA. Inosine also helps reduce immune reactions to therapeutic RNAs by preventing detection by sensors like MDA5, improving their safety. Its ability to pair flexibly with different bases allows better control over RNA structure and protein production, which is useful in designing therapies like antisense, siRNA, and mRNA. However, challenges such as unintended edits, variable efficiency, and delivery issues still limit clinical translation.

Glycerol nucleic acids (GNA) Glycol nucleic acid (GNA) is a nucleic acid analogue featuring an acyclic three-carbon glycol backbone that can still support Watson–Crick base pairing and duplex formation, despite lacking the conventional ribose sugar found in DNA and RNA (Zhang et al., 2005). In the context of siRNA therapeutics, GNA is incorporated as a chemical modification to improve pharmacological properties, particularly safety. Specifically, (S)-GNA substitutions—most effectively at position 7 of the antisense strand seed region—introduce controlled thermodynamic destabilization that reduces unintended, seed-mediated off-target interactions while preserving on-target gene silencing (Schlegel et al., 2022). This modification also enhances nuclease resistance and can improve strand selection and loading into the RNA-induced silencing complex (RISC), thereby maintaining or even increasing potency (Schlegel et al., 2017). Overall, GNA is used in siRNA therapeutics because it enables fine-tuning of duplex stability and structure, leading to improved specificity, reduced toxicity, and favorable clinical performance.



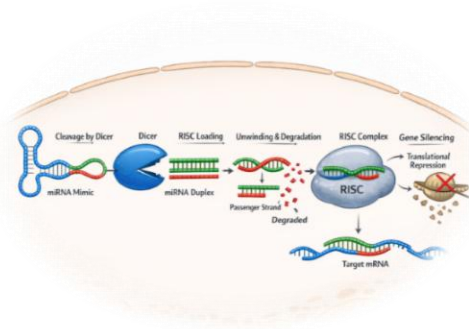
Elebsiran (ALN-HBV02), a GalNAc-conjugated siRNA developed for chronic hepatitis B uses the “enhanced stabilization chemistry” (ESC+), within which a GNA substitution contributes to improved strand specificity and reduced off-target activity.

In Phase 1/2 studies, Elebsiran demonstrated knockdown of HBsAg, with reductions often exceeding 1.5–2.0 log₁₀ IU/mL following subcutaneous dosing at 300 mg every 4–8 weeks, and evidence of sustained activity beyond the dosing interval (Yuen et al., 2022; Gane et al., 2023). Importantly, the safety profile was favorable, with no dose-limiting toxicities and a low incidence of ALT elevations, supporting the hypothesis that chemical refinements such as GNA substitutions mitigate hepatocellular stress linked to off-target effects. Across cohorts, repeated dosing maintained efficacy without evidence of tachyphylaxis, and pharmacodynamic responses were consistent with prolonged hepatic RISC loading. (Yuen et al., 2022; Gane et al., 2023).

Design Strategies

miRNA mimics

A microRNA (miRNA) mimic is typically designed as a synthetic double-stranded RNA that resembles the endogenous miRNA duplex. It contains a guide strand (Antisense) corresponding to the mature miRNA sequence and a complementary passenger strand (Sense), forming a short duplex with characteristic features such as 2-nucleotide 3' overhangs generated by Dicer processing.



After introduction into the cell, the miRNA mimic duplex is unwound during RISC loading, the passenger strand is degraded, and the guide strand is incorporated into the RNA-induced silencing

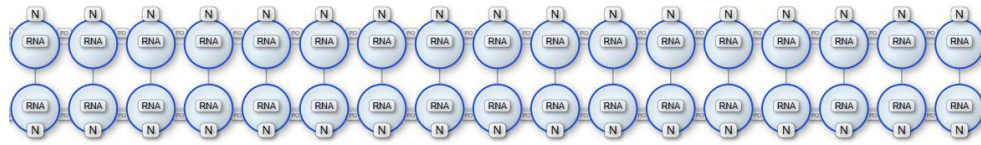
complex (RISC). Within RISC, the guide strand directs gene silencing by partial base pairing with target mRNAs, leading to translational repression or mRNA degradation.

MRX34 was a first-in-class microRNA therapeutic composed of a synthetic double-stranded miR-34a mimic (UGGCAGUGUCUUAGCUGGUUGU) formulated within a lipid nanoparticle delivery system to restore the tumor-suppressive miR-34 pathway downstream of p53 and repress multiple oncogenic targets. Preclinical studies and early Phase 1 clinical data demonstrated successful systemic delivery, target engagement, and limited antitumor activity, supporting proof-of-concept for miRNA replacement therapy. However, the trial was terminated in 2016 due to severe immune-related toxicities, including cytokine release syndrome and treatment-related deaths, likely caused by innate immune activation triggered by the nanoparticle. This outcome highlighted key challenges in RNA therapeutics particularly the balance between structural design, delivery efficiency, and immunogenicity despite the strong biological rationale for miR-34-based therapy (Zhang et al., 2019; Hong et al., 2020)

MRX34 is an unmodified RNA formulated using a liposomal nanoparticle delivery system (NOV340) composed of amphoteric lipids designed to encapsulate and protect the double-stranded miR-34a mimic during systemic administration. The lipid formulation typically included components such as phospholipids, cholesterol, and PEGylated lipids to enhance stability, prolong circulation time, and facilitate cellular uptake and endosomal release of the RNA cargo.

In the Phase 1 clinical trial, MRX34 was administered intravenously twice weekly (commonly on days 1 and 4 of a 3-week cycle), with dose escalation up to a maximum tolerated dose of approximately 110 mg/m² in non-hepatocellular carcinoma patients and 93 mg/m² in HCC patients, before immune-related toxicities limited further dosing (Hong et al., 2020; Beg et al., 2017). The toxicity and clinical outcomes observed with MRX34 cannot be attributed to a single factor, as multiple confounding elements were involved. The lipid nanoparticle delivery system facilitated broad systemic distribution, potentially exposing diverse tissues to the therapy, while frequent and high dosing likely led to excessive off-target accumulation. The complexity of the drug product complicates efforts to determine whether toxicity was primarily driven by the delivery system, payload miRNA, or the intrinsic biology of miRNA regulation.

siRNA



Length and Structure

21–23 nucleotides (Elbashir et al., 2001)

2-nt 3' overhangs (usually “UU” or “TT”) (Elbashir et al., 2001)

Double-stranded RNA duplex required for RISC loading (Sioud, 2025)

Strand Selection and Thermodynamic Asymmetry

Lower stability at the 5' end of antisense (guide) strand (Schwarz et al., 2003)

Higher stability at the 5' end of sense (passenger) strand (Schwarz et al., 2003)

This promotes correct guide strand loading into RISC (Sioud, 2025)

A/U-rich at antisense 5' end (Ui-Tei et al., 2004)

G/C-rich at sense 5' end (Ui-Tei et al., 2004)

GC Content

Optimal ~30–50% GC (Reynolds et al., 2004)

Too high GC leads to overly stable duplex and poor RISC unwinding (Reynolds et al., 2004)

Too low GC results in weak target binding (Reynolds et al., 2004)

Functional siRNAs tend to have moderate GC and lower internal stability (Sioud, 2025)

Nucleotide Preferences (Empirical Rules)

Antisense strand:

A/U at position 1 (5' end) (Ui-Tei et al., 2004)

Sense strand:

G/C at 5' end (Ui-Tei et al., 2004)

Seed region (positions 2–8):

Prefer A/U-rich (Reynolds et al., 2004)

Avoid long GC stretches or repeats (Reynolds et al., 2004)

Seed Region Considerations (Positions 2–8)

Critical for target recognition (Sioud, 2025)

Main source of off-target effects (Jackson and Linsley, 2010)

Partial complementarity can silence unintended genes (miRNA-like effects) (Jackson and Linsley, 2010)

Avoid seed matches in:

3' UTRs of unrelated genes (Jackson and Linsley, 2010)

Chemical modifications (e.g. 2'-O-methyl) can reduce off-target effects (Lennox and Behlke, 2016)

Target Site Considerations

- Prefer accessible regions with low secondary structure

- Avoid:

 - Start codon region (Reynolds et al., 2004)

 - Highly structured regions

Avoid Undesirable Motifs

- Avoid long repeats and palindromes (Reynolds et al., 2004)

- Avoid internal hairpins (Reynolds et al., 2004)

- Avoid immune-stimulatory motifs (e.g. GU-rich) (Sioud, 2025)

Overhang Design

- Standard 2-nt 3' overhangs (Elbashir et al., 2001)

- Asymmetric overhangs (antisense favored) can improve potency (Sano et al., 2008; Sioud, 2025)

Off-Target Minimization

- Design multiple siRNAs (≥ 3 per gene) (Sioud, 2025)

- Modify seed region if needed (Jackson and Linsley, 2010)

- Avoid perfect matches to unintended transcripts (Jackson and Linsley, 2010)

Gapmer (RNase-H) ASOs

Structure and Length

- Single-stranded oligonucleotides (Crooke, 2017)

- Typically 16–20 nucleotides long (Bennett and Swayze, 2010)

- Gapmer design consists of:

 - Central DNA gap (≈ 8 –10 nt) required for RNase H activity (Crooke, 2017)

 - Modified flanking regions (“wings”) to enhance stability and binding (Lennox and Behlke, 2016)

Mechanism Consideration (RNase H Activation)

- Gap must be composed of DNA nucleotides

- Enables formation of RNA–DNA hybrid (Bennett and Swayze, 2010)

- Recruits RNase H, which cleaves the target RNA (Crooke, 2017)

Chemical Modifications

- Essential for stability, affinity, and safety (Lennox and Behlke, 2016)

- Common wing modifications:

 - Locked nucleic acids (LNA) (Kurreck, 2003)

 - 2'-O-methyl (2'-OMe) (Lennox and Behlke, 2016)

 - 2'-O-methoxyethyl (2'-MOE) (Bennett and Swayze, 2010)

- Backbone modification:

 - Phosphorothioate (PS) backbone (standard) (Geary et al., 2015)

Improves:

- Nuclease resistance (Geary et al., 2015)
- Plasma protein binding (Geary et al., 2015)
- Cellular uptake (Geary et al., 2015)

GC Content and Binding Affinity

- Optimal GC content ~40–60% (Bennett and Swayze, 2010)
- Must ensure strong hybridization to target RNA (Crooke, 2017)
- Avoid excessively high GC due to toxicity and off-target binding (Lennox and Behlke, 2016)

Target melting temperature (T_m):

- Typically ~55–65°C (context-dependent) (Watts and Corey, 2012)

Target Site Selection

Prefer accessible regions with low RNA secondary structure (Watts and Corey, 2012)

Can target:

- mRNA (coding region or UTRs) (Bennett and Swayze, 2010)
- Pre-mRNA (intronic sequences) (Crooke, 2017)

Avoid:

- Highly structured regions (Watts and Corey, 2012)
- Protein-bound regions (Crooke, 2017)
- Repetitive sequences (Lennox and Behlke, 2016)

Avoid Undesirable Sequence Motifs

- Avoid CpG motifs (immune stimulation) (Krieg, 2006)
- Avoid long G runs (e.g. GGGG → aggregation/toxicity) (Lennox and Behlke, 2016)
- Avoid palindromic sequences (Kurreck, 2003)
- Avoid self-complementarity (hairpin formation) (Kurreck, 2003)

Off-Target Minimization

- Ensure high specificity to target RNA (Crooke, 2017)
- Avoid partial complementarity to unintended transcripts (Watts and Corey, 2012)
- Perform transcriptome-wide homology screening (e.g. BLAST) (Bennett and Swayze, 2010)
- Design multiple candidate ASOs (≥3–5 per target) (Crooke, 2017)

Toxicity Considerations

- Monitor sequence-dependent toxicity (Lennox and Behlke, 2016)
- Monitor off-target RNase H cleavage (Crooke, 2017)
- Certain motifs (e.g. CpG, high GC stretches) increase risk (Krieg, 2006)

Cellular Localization

Gapmers are effective in:

- Nucleus (pre-mRNA targeting) (Crooke, 2017)
- Cytoplasm (Bennett and Swayze, 2010)

No requirement for RISC loading (Crooke, 2017)

RNase-H considerations

RNase H is a conserved enzyme that maintains genome stability by removing RNA–DNA hybrids (R-loops) formed during transcription. If these hybrids accumulate, they can cause DNA damage such as double-stranded breaks and chromosomal instability. RNase H1 and RNase H2 catalytically degrade the RNA component of these structures, acting as part of a cellular surveillance system that prevents harmful hybrid buildup and protects genomic integrity (Wahba et al., 2011).

Oligonucleotide backbones can be systematically tuned by replacing a fully phosphorothioate (PS) backbone with a small number of phosphodiester (PO) linkages to reduce charge and protein binding while retaining function. By introducing PO linkages stepwise (1–3 positions) in defined locations, nuclease stability decreases gradually rather than abruptly, and RNase H cleavage activity is largely preserved if sufficient PS content remains within the enzyme's binding footprint. The key insight is that backbone composition can be optimized; not all PS is required so long as modifications are limited and strategically placed to maintain the structural interactions necessary for RNase H activity. Common clinically approved gapmer strategies for reducing PS content but maintaining activity illustrate this: soooooossssssssooss.

Steric-Blocking single-stranded ASOs

Structure and Length

Single-stranded oligonucleotides 15-25 nt in length (Bennett and Swayze, 2010)

Fully modified backbone (**no DNA gap**) since RNase H is not required (Lennox and Behlke, 2016)

Mechanism Consideration (Steric Blocking)

Block molecular interactions such as:

- Ribosome binding (translation inhibition)

- Spliceosome assembly (splice modulation)

- Protein–RNA interactions (Watts and Corey, 2012)

Chemical Modifications

Fully modified nucleotides required for stability and affinity (Lennox and Behlke, 2016)

Common chemistries:

- 2'-O-methyl (2'-OMe) (Lennox and Behlke, 2016)

- 2'-O-methoxyethyl (2'-MOE) (Bennett and Swayze, 2010)

- Locked nucleic acids (LNA) (Kurreck, 2003)

- Phosphorodiamidate morpholino oligomers (PMOs) (Summerton, 2007)

Backbone:

- Phosphorothioate (PS) often used (Geary et al., 2015)

- PMOs use neutral backbone (Summerton, 2007)

Purpose:

- Increase binding affinity

- Improve nuclease resistance
- Enhance cellular uptake

Binding Affinity and GC Content

- Moderate GC content preferred (~40–60%) (Watts and Corey, 2012)
- High affinity is critical since no enzymatic amplification occurs (Crooke, 2017)
- Avoid excessively high GC to reduce off-target binding and toxicity (Lennox and Behlke, 2016)

Target Site Selection

Common target regions:

- Translation start site (AUG region)
- Splice sites (exon–intron junctions)
- Splicing enhancers/silencers (ESE/ESS) (Bennett and Swayze, 2010)

Prefer:

- Accessible regions with low secondary structure (Watts and Corey, 2012)

Avoid:

- Highly structured RNA regions
- Protein-bound regions

Splice Modulation (if applicable)

Can induce:

- Exon skipping
- Exon inclusion (Bennett and Swayze, 2010)

Requires precise targeting of splice regulatory elements

Avoid Undesirable Sequence Motifs

- Avoid CpG motifs (immune stimulation) (Krieg, 2006)
- Avoid long G runs (aggregation/toxicity) (Lennox and Behlke, 2016)
- Avoid palindromes and self-complementarity

Off-Target Minimization

Complicated to assess and context specific.

Toxicity Considerations

- Monitor sequence-dependent toxicity (Lennox and Behlke, 2016)
- Monitor immune activation (Krieg, 2006)
- High-affinity chemistries (e.g. LNA) may increase toxicity if overused

Cellular Localization

- Can function in both nucleus and cytoplasm (Crooke, 2017)
- Splice-switching ASOs act primarily in the nucleus

Experimental Validation

- Confirm functional effect (e.g. splice change or translation block) (Bennett and Swayze, 2010)
- Confirm specificity and minimal off-target effects
- Confirm low cytotoxicity

Compare Splicing Technology: PMO vs 2'-MOE

FEATURE	PMO (MORPHOLINO; E.G., ETEPLIRSEN, GOLODIRSEN, VILTOLARSEN)	2'-MOE PS ASO (E.G., NUSINERSEN)
Backbone	Phosphorodiamidate (neutral)	Phosphorothioate (negatively charged)
Sugar/Scaffold	Morpholine ring (no ribose)	Ribose with 2'-MOE modification
Charge	Neutral	Negative
Protein binding	Low	High
Nuclease resistance	Very high	High
Binding affinity (Tm)	High	High–very high
Cellular uptake	Poor (often needs high dose or conjugates)	Better (PS-driven uptake)
Distribution	Limited without conjugates	Broad tissue distribution
Toxicity profile	Generally low	Moderate (PS-related effects possible)
Mechanism	Steric blocking (splice modulation)	Steric blocking (splice modulation)

Design implications: - PMOs prioritize **safety and stability**, with tradeoffs in delivery efficiency. - MOE-PS ASOs balance **delivery, potency, and pharmacokinetics**, with increased biological interactions.

Approach	Key Developers	Technology
Nonselective ASO (total HTT lowering)	Roche + Ionis Pharmaceuticals	2'-MOE gapmer antisense oligonucleotides with mixed PS/PO backbone (stereorandom)
Allele-selective ASO (SNP-targeted)	Wave Life Sciences	2'-O-Me gapmer antisense oligonucleotides with stereopure synthesis
Next-generation allele-selective ASO	Wave Life Sciences	PS/PO/PN backbone ASOs (phosphoramidate-modified for improved distribution and potency)
RNA interference (RNAi)	Academic / preclinical groups	siRNA or shRNA delivered via viral vectors
General ASO platform	Multiple groups	Chemically modified single-stranded DNA oligonucleotides targeting RNA

Mechanism of Action	Advantages	Limitations / Outcomes
RNase H-mediated degradation of both mutant and wild-type HTT mRNA after ASO binding	Broad applicability across all patients; robust and dose-dependent HTT lowering	Loss of wild-type HTT function may cause toxicity; clinical trials halted due to safety concerns and lack of efficacy
Selective binding to SNP-linked mutant HTT mRNA → RNase H degradation of mutant transcript only	Preserves wild-type HTT; reduces risk of on-target toxicity	Limited to patients with specific SNPs; insufficient target engagement in trials; programs halted
RNase H-mediated degradation with improved pharmacokinetics and distribution	Enhanced stability, potency, and CNS distribution (preclinical improvements)	Still early-stage; clinical efficacy and safety not yet established
RISC-mediated cleavage of HTT mRNA via siRNA/shRNA pathways	Strong and sustained gene silencing in preclinical models	Delivery challenges; less controllable dosing; primarily preclinical

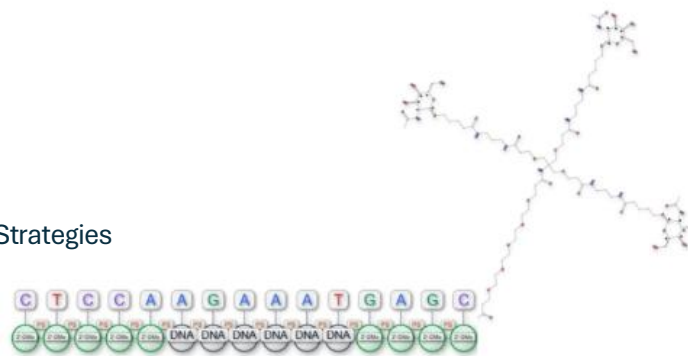
**Hybridization to RNA
(mRNA/pre-mRNA)
enabling degradation or
modulation of RNA
processing**

Flexible design; broad CNS
distribution; no viral vector
required

Requires repeated dosing; potential off-target effects
and immune activation

Conjugates and Delivery Strategies

GalNAc Conjugation



The development of GalNAc-based liver targeting therapeutics stems from the discovery of the asialoglycoprotein receptor (ASGPR) in the 1960s by Ashwell and Morell, who showed that hepatocytes selectively clear desialylated glycoproteins via recognition of exposed terminal galactose and N-acetylgalactosamine (GalNAc) (Ashwell and Morell, 1968; Morell et al., 1966). Subsequent studies established that receptor binding is enhanced by multivalency, laying the foundation for synthetic ligand design. By optimizing ligand valency, linker design, and synthesis, Dr. Muthiah Manoharan (Alnylam) and colleagues developed the first triantennary GalNAc ligands that exist in the clinic today (Nair et al., 2014; Egli and Manoharan, 2023).

Lipid and Liposomal Systems

Despite their utility, LNPs are immunologically active systems, and their safety profile is tightly linked to their lipid composition and biodistribution. Chronic dosing regimens can lead to accumulation and safety issues. A major development is the introduction of biodegradable ionizable lipids, which break down into non-toxic metabolites after delivery, reducing hepatic accumulation and chronic inflammatory responses. Improved lipid chemistries are to minimize activation of innate immune pathways, lowering cytokine release.

Patisiran (Onpattro), an LNP-formulated RNAi therapeutic, established that chronic systemic delivery of siRNA is clinically feasible. However, its safety and administration burden reflect the inherent liabilities of lipid nanoparticles. In the APOLLO trial, patients treated with patisiran experienced higher rates of infusion-related reactions (19% vs 9% with placebo) and peripheral edema (30% vs 22%), along with systemic adverse events such as fatigue, diarrhea, and dizziness. These effects are consistent with known LNP-associated mechanisms, including complement activation, innate immune stimulation, and vascular responses. As a result, patisiran requires intravenous infusion and routine premedication (e.g., corticosteroids, antihistamines) to mitigate hypersensitivity and infusion reactions.

In contrast, vutrisiran is a next-generation RNAi therapeutic targeting the same gene (TTR) but uses a GalNAc conjugate rather than LNPs. This enables direct hepatocyte uptake via the asialoglycoprotein receptor, avoiding many of the systemic immune interactions associated with nanoparticle carriers. Consequently, vutrisiran can be administered as a subcutaneous injection once every three months, substantially reducing treatment burden.

Vutrisiran demonstrates a distinct and generally improved tolerability profile. In the HELIOS-A study, the most common adverse events in patients with hATTR-PN were pain in extremity (15%), arthralgia (11%), dyspnea (7%), and decreased vitamin A levels (7%), with no signal of infusion-related reactions. Similarly, in ATTR-CM, no new safety concerns were identified. Notably, unlike patisiran, vutrisiran does not require premedication, underscoring the advantage of eliminating LNP-associated immune activation and enabling a more convenient, repeat dosing paradigm.

PEGylation

PEGylation, the covalent attachment of polyethylene glycol (PEG) to therapeutic molecules. The technology was discovered in the 1960s by Frank Davis who proposed its use to reduce the immunogenicity of protein drugs by shielding them from immune recognition and prolonging circulation time (Veronese & Pasut, 2005; Pasut & Veronese, 2012). This concept was experimentally validated in a landmark study by Abuchowski et al. (1977), which demonstrated that PEGylation of bovine liver catalase significantly decreased immunogenicity while extending plasma half-life.

PEGylation transitioned from academic concept to a commercial technology (Harris & Chess, 2003) with the FDA approval of Adagen in 1990, followed by Oncaspar in 1994. (Alconcel et al., 2011; Harris & Chess, 2003). From the mid-1990s onward, PEGylation rapidly expanded across diverse drug classes and architectures, including the development of PEGylated liposomes such as Doxil (Allen & Cullis, 2013), site-specific mono-PEGylated proteins like Pegintron, and branched PEG systems exemplified by Pegasys (Veronese & Mero, 2008).

The technology further diversified to include PEGylated aptamers (Macugen™), antibody fragments (Cimzia™), peptides (Omontys), and small molecules (Movantik), reflecting its broad applicability across therapeutic modalities (Keefe et al., 2010; Fishbane et al., 2013). More recently, PEGylation has played a critical role in nanomedicine, particularly in lipid nanoparticle (LNP) systems used for nucleic acid delivery; a critical component of the Onpattro LNP and the widespread deployment of PEGylated LNP-based mRNA COVID-19 vaccines (Cullis & Hope, 2017; Hou et al., 2021).

While there are many more ligand conjugate technology used including palmitate and antibodies the vast majority of pro-drug technologies for tissue specificity is for liver with various forms of GalNAc.

RNA-targeting clinical review

Several trends define the next generation of oligonucleotide therapeutics:

- Increasing chemical diversity (beyond PS and standard 2' modifications)
- Greater use of noncanonical bases
- Integration of multiple backbone types within a single molecule
- Precision structural engineering for enzyme recruitment
- Continued dominance of GalNAc for liver delivery

COMPOUND	BASE	BACKBONE	SUGAR	CONJUGATION	ROUTE
FOMIVIRSEN	21-mer antisense DNA	Full phosphorothioate	DNA	None	Intravitreal injection
MIPOMERSEN	Gapmer ASO	Phosphorothioate	2'-MOE + DNA gap	None	Subcutaneous injection
ETEPLIRSEN	30-mer antisense	PMO	Morpholino	None	Intravenous infusion
NUSINERSEN	18-mer	Phosphorothioate	2'-MOE	None	Intrathecal injection
INOTERSEN	Gapmer ASO	Phosphorothioate	2'-MOE	None	Subcutaneous injection
GOLODIRSEN	Antisense	PMO	Morpholino	None	Intravenous infusion
VOLANESORSEN	Gapmer ASO	Phosphorothioate	2'-MOE	None	Subcutaneous injection
VILTOLARSEN	Antisense	PMO	Morpholino	None	Intravenous infusion
CASIMERSEN	Antisense	PMO	Morpholino	None	Intravenous infusion
TOFERSEN	20-mer gapmer	Mixed PS/PO	2'-MOE	None	Intrathecal injection
EPLONTERSEN	Gapmer ASO	Mixed PS/PO	2'-MOE	GalNAc	Subcutaneous injection
PEGAPTANIB	RNA aptamer	Modified RNA	Modified ribose	PEG	Intravitreal injection

AVACINCAPTAD PEGOL	RNA aptamer	Modified RNA	Modified ribose	PEG	Intravitreal injection
PATISIRAN	siRNA	Phosphodiester	2'-OMe	LNP	Intravenous infusion
GIVOSIRAN	siRNA	PS modified	2'-F/2'-OMe	GalNAc	Subcutaneous injection
LUMASIRAN	siRNA	PS modified	2'-F/2'-OMe	GalNAc	Subcutaneous injection
INCLISIRAN	siRNA	PS modified	2'-F/2'-OMe	GalNAc	Subcutaneous injection
VUTRISIRAN	siRNA	PS enhanced	2'-F/2'-OMe	GalNAc	Subcutaneous injection
NEDOSIRAN	siRNA	PS/PO	2'-F/2'-OMe	GalNAc	Subcutaneous injection
MIRAVIRSEN	Anti-miR	Phosphorothioate	LNA/DNA	None	Subcutaneous injection
RG-101	Anti-miR	Modified ASO	Not specified	GalNAc	Subcutaneous injection

DNA = deoxyribose

2'-MOE = 2'-O-methoxyethyl

2'-OMe = 2'-O-methyl

2'-F = 2'-fluoro

LNA = locked nucleic acid

PMO = morpholino backbone

Institutions by Platform

This section maps the major chemistries described above to leading companies and institutions that have developed or commercialized them.

siRNA Platforms (2'-OMe / 2'-F + PS)

Alnylam Pharmaceuticals – Active

Arrowhead Pharmaceuticals – Active

Dicerna – Active (integrated into Novo Nordisk)

Silence Therapeutics – Active

Gapmer ASOs (MOE / OMe / DNA gap + PS)

Ionis Pharmaceuticals – Active

Biogen – Active

Roche / Genentech – Active

Steric-Blocking ASOs (Fully Modified)

Sarepta Therapeutics – Active

Nippon Shinyaku – Active

Avidity Biosciences – Acquired by Novartis

Biogen / Ionis

RNA Editing Oligos (ADAR / EON Platforms)

ProQR Therapeutics – Active

Wave Life Sciences – Active

Beam Therapeutics – Active

Academic institutions – Active (various universities and research institutes)

Alternative Chemistries and Platforms

Eyetech / Pfizer (Pegaptanib) – Legacy (Eyetech acquired; program commercialized historically)

Phio Pharmaceuticals – Active

Wave Life Sciences (stereopure PS) – Active

Industry Activity Status

This supplemental section provides a consolidated view of organizations referenced in this white paper and their current operational status in the oligonucleotide therapeutics field.

Company / Institution	Primary Chemistry Focus	Notes
Alnylam Pharmaceuticals	siRNA (2'-OMe / 2'-F, GalNAc)	Market leader in RNAi therapeutics
Arrowhead Pharmaceuticals	siRNA (TRiM platform)	Expanding beyond liver delivery
Dicerna (Novo Nordisk)	siRNA (GalXC)	Fully integrated into Novo Nordisk
Silence Therapeutics	siRNA	Focus on mRNAi GOLD platform
Ionis Pharmaceuticals	Gapmer ASO (MOE, cEt)	Pioneer in antisense chemistry
Biogen	ASO (partnered programs)	CNS-focused oligonucleotide drugs
Roche / Genentech	ASO / RNA therapeutics	Broad RNA therapeutic portfolio
Sarepta Therapeutics	PMO ASO	Leader in Duchenne exon skipping
Nippon Shinyaku	PMO ASO	Partner in DMD therapies
Avidity Biosciences	Conjugated ASOs	Antibody-oligo conjugates
ProQR Therapeutics	RNA editing (Axiomer)	Focus on ADAR-mediated editing
Wave Life Sciences	Stereopure ASO / editing	Chirality-controlled PS chemistry
Beam Therapeutics	RNA/base editing	Expanding into RNA editing
Phio Pharmaceuticals	Self-delivering RNAi	Immuno-oncology focus
Eyetech (Pegaptanib originator)	Aptamer	Acquired; program commercialized historically
Academic Institutions	Various (ASO, editing)	Key drivers of early innovation

References

- Matsukura M, Zon G, Shinozuka K, et al. Inhibition of HIV replication by antisense oligodeoxynucleotides. *Proc Natl Acad Sci U S A*. 1987;84:7706–7710.
- Østergaard ME, De Hoyos CL, Wan WB, Shen W, Low A, Berdeja A, et al. Understanding the effect of controlling phosphorothioate chirality in the DNA gap on the potency and safety of gapmer antisense oligonucleotides. *Nucleic Acids Res*. 2020;48(4):1691–1700.
- Bennett CF, Baker BF, Pham N, Swayze E, Geary RS. Pharmacology of antisense drugs. *Annu Rev Pharmacol Toxicol*. 2017;57:81–105.
- Eckstein F. Phosphorothioates, essential components of therapeutic oligonucleotides. *Nucleic Acid Ther*. 2014;24(6):374–387.
- Prakash TP, Siwkowski A, Allerson CR, Vasquez G, Lee S, Wancewicz EV, et al. Short antisense oligonucleotides with conformationally restricted nucleoside analogues show improved potency without increased toxicity. *J Med Chem*. 2009;52(1):10–13.
- Geary RS, Norris D, Yu R, Bennett CF. Pharmacokinetics, biodistribution and cell uptake of antisense oligonucleotides. *Adv Drug Deliv Rev*. 2015;87:46–51.
- Lovrić J, Yan J, Li XQ, Karlsborn T, Bood M, Dahlén A, et al. In vitro structure–activity relationship stability study of antisense oligonucleotide therapeutics using biological matrices and nucleases. *Pharmacol Res Perspect*. 2025;13:e70096.
- Zanardi, T. A., Kim, T.-W., Shen, L., Serota, D., Papagiannis, C., Park, S.-Y., Kim, Y., & Henry, S. P. (2018). Chronic toxicity assessment of 2'-O-methoxyethyl antisense oligonucleotides in mice. *Nucleic Acid Therapeutics*, 28(4), 233–240.
- Zanardi, T. A., Kim, T.-W., Shen, L., et al. (2018). Chronic toxicity assessment of 2'-O-methoxyethyl antisense oligonucleotides in mice. *Nucleic Acid Therapeutics*, 28(4), 233–240.
- Kurreck, J., Wyszko, E., Gillen, C., and Erdmann, V. A. (2002). Design of antisense oligonucleotides stabilized by locked nucleic acids. *Nucleic Acids Res.*, 30(9), 1911–1918.
- Aartsma-Rus, A., Corey, D. R., and others (2023). The evolving landscape of antisense oligonucleotide therapeutics. *Nat Rev Drug Discov.*, 22, 1–22.
- Geary, R. S., et al. (2015). Pharmacokinetics, biodistribution and cell uptake of antisense oligonucleotides. *Adv Drug Deliv Rev.*, 87, 46–51.
- Crooke, S. T., et al. (2018). Antisense drug discovery and development technology considered in a pharmacological context. *Biochemical Pharmacology*, 153, 17–28.
- Crooke, S. T., Witztum, J. L., Bennett, C. F., & Baker, B. F. (2017). Antisense drug discovery and development technology. *Nat Rev Drug Discov.*, 16(6), 401–423.

Souleimanian, N., Deleavey, G.F., Soifer, H., Wang, S., Tiemann, K., Damha, M.J., & Stein, C.A. (2012). Antisense 2'-deoxy-2'-fluoroarabino nucleic acid (2'F-ANA) oligonucleotides: In vitro gymnotic silencers of gene expression. *Molecular Therapy – Nucleic Acids*.

Rüger, J., Ioannou, S., Castanotto, D. and Stein, C.A., 2020. Oligonucleotides to the (Gene) Rescue: FDA approvals 2017–2019. *Trends in Pharmacological Sciences*, 41(1), pp.27–41.

Clinical RNAi & Approved Therapies

Adams D, Gonzalez-Duarte A, O’Riordan WD, et al. Patisiran, an RNAi therapeutic, for hereditary transthyretin amyloidosis. *N Engl J Med*. 2018;379:11–21.

Obici L, Berk JL, González-Duarte A, et al. Quality of life outcomes in APOLLO study of patisiran. *Amyloid*. 2020;27(3):153–162.

U.S. Food and Drug Administration. ONPATTRO (patisiran) [prescribing information]. Silver Spring, MD: FDA; 2018.

Adams D, Tournev IL, Taylor MS, et al. Efficacy and safety of vutrisiran in hereditary transthyretin-mediated amyloidosis (HELIOS-A). *Amyloid*. 2023;30(2):73–82.

Obici L, Ajroud-Driss S, Lin KP, et al. Impact of vutrisiran on quality of life and physical function in hereditary transthyretin-mediated amyloidosis. *Neurol Ther*. 2023;12:1–15.

Habtemariam BA, Karsten V, Attarwala H, et al. Pharmacokinetics and pharmacodynamics of vutrisiran. *Clin Pharmacol Ther*. 2021;110(3):702–713.

U.S. Food and Drug Administration. AMVUTTRA (vutrisiran) [prescribing information]. Silver Spring, MD: FDA; 2022.

Allen TM, Cullis PR. Liposomal drug delivery systems: from concept to clinical applications. *Adv Drug Deliv Rev*. 2013;65(1):36–48.

Webster LR, et al. Randomized clinical trial of naloxegol for opioid-induced constipation. *J Pharmacol Exp Ther*. 2013;347(2):408–416.

Adams D, et al. Patisiran, an RNAi therapeutic, for hereditary transthyretin amyloidosis. *N Engl J Med*. 2020;379:11–21.

Garrelfs SF, et al. Lumasiran, an RNAi therapeutic for primary hyperoxaluria type 1. *N Engl J Med*. 2021;384:1216–1226.

Raal FJ, et al. Inclisiran for the treatment of heterozygous familial hypercholesterolemia. *N Engl J Med*. 2020;382:1520–1530.

Santaris Pharma A/S (2012). Phase 2a clinical study of miravirsin (SPC3649) in treatment-naïve patients with chronic hepatitis C virus infection. Clinical study reports and summaries.

Seth, P. P., et al. (2010). Second-generation antisense oligonucleotides: design, mechanism, and clinical application. *Current Opinion in Molecular Therapeutics*, 12(3), 289–298.

Khvorova, A., & Watts, J. K. (2017). The chemical evolution of oligonucleotide therapies of clinical utility. *Nature Biotechnology*, 35, 238–248.

Drygin, D. (2022). Lessons Learned from Preclinical and Clinical Development of microRNA Therapies. RNA Leaders World Congress, Basel.

König, J., Nies, A. T., Cui, Y., Leier, I., & Keppler, D. (1999). A novel human canalicular multispecific organic anion transporter (MRP2). *Journal of Clinical Investigation*, 104(2), 167–174.

European Medicines Agency. Leqvio (inclisiran) EPAR: assessment report. Amsterdam: EMA; 2020.

U.S. Food and Drug Administration. Pharmacology/Toxicology Review: Inclisiran (NDA 214012). Silver Spring, MD: FDA; 2021.

AstraZeneca. 2024. AZD5312 clinical development updates. AstraZeneca internal and public pipeline reports.

ClinicalTrials.gov. Study of AZD5312 in metastatic castration-resistant prostate cancer. Bethesda, MD: U.S. National Library of Medicine; 2025. Available from: <https://clinicaltrials.gov>.

de Bono, J., et al. 2023. “Targeting androgen receptor signaling with antisense oligonucleotides in prostate cancer.” *Journal of Clinical Oncology* (or relevant conference proceeding).

Zhang, L., Liao, Y., & Tang, L. (2019). MicroRNA-34 family: a potential tumor suppressor and therapeutic candidate in cancer. *Journal of Experimental & Clinical Cancer Research*

CADTH (2020) Clinical Review Report: Inotersen (Tegsedi). Canadian Agency for Drugs and Technologies in Health

Schlegel, M.K., Janas, M.M., Jiang, Y., Barry, J.D., Davis, W., Agarwal, S., Berman, D., Brown, C.R., Castoreno, A., LeBlanc, S. and Manoharan, M., 2022. From bench to bedside: improving the clinical safety of GalNAc-siRNA conjugates using seed-pairing destabilization. *Nucleic Acids Research*, 50(12), pp.6656–6670.

Yuen, M.-F., Locarnini, S., Lim, Y.-S., Strasser, S.I., Sievert, W., Cheng, W., Subramanian, G.M., Myers, R.P., and Gane, E. (2022) ‘Elebsiran, an RNA interference therapeutic targeting hepatitis B virus, demonstrates sustained HBsAg reduction in chronic hepatitis B patients’, *Hepatology*, 76(5), pp. 1231–1242.

FDA (2020) Integrated Review: Oxlumo (lumasiran), NDA 214103. U.S. Food and Drug Administration.

Delivery Systems (LNPs, GalNAc, etc.)

Tenchov R, Sasso JM, Zhou Q. Lipid nanoparticles—from liposomes to mRNA vaccine delivery. *ACS Nano*. 2021;15(11):16982–17015.

Hou X, Zaks T, Langer R, Dong Y. Lipid nanoparticles for mRNA delivery. *Nat Rev Mater*. 2021;6:1078–1094.

Veronese FM, Pasut G. PEGylation, successful approach to drug delivery. *Drug Discov Today*. 2005;10(21):1451–1458.

Cullis PR, Hope MJ. Lipid nanoparticle systems for enabling gene therapies. *Mol Ther*. 2017;25(7):1467–1475.

Hou X, Zaks T, Langer R, Dong Y. Lipid nanoparticles for mRNA delivery. *Nat Rev Mater*. 2021;6:1078–1094.

Qassem, S., Breier, D., Naidu, G. S., Hazan-Halevy, I., & Peer, D. (2024). Unlocking the therapeutic potential of locked nucleic acids through lipid nanoparticle delivery. *Molecular Therapy: Nucleic Acids*, 35, 1–12.

Geary, R.S., Norris, D., Yu, R. and Bennett, C.F. (2015) 'Pharmacokinetics, biodistribution and cell uptake of antisense oligonucleotides', *Advanced Drug Delivery Reviews*, 87, pp. 46–51.

Springer, A.D. and Dowdy, S.F., 2018. GalNAc-siRNA conjugates: leading the way for delivery of RNAi therapeutics. *Nucleic Acid Therapeutics*, 28(3), pp.109–118.

Chemical Modifications & ASO Chemistry

Pasut G, Veronese FM. State of the art in PEGylation: the great versatility achieved after forty years of research. *J Control Release*. 2012;161(2):461–472.

Harris JM, Chess RB. Effect of PEGylation on pharmaceuticals. *Nat Rev Drug Discov*. 2003;2(3):214–221.

Veronese FM, Mero A. The impact of PEGylation on biological therapies. *BioDrugs*. 2008;22(5):315–329.

Seth PP, Swayze EE. The medicinal chemistry of RNase H-activating antisense oligonucleotides. In: Wan Y, editor. *Advances in Nucleic Acid Therapeutics*. Cambridge: Royal Society of Chemistry; 2018. p. 32–61.

Wan WB, Seth PP. The medicinal chemistry of therapeutic oligonucleotides. *J Med Chem*. 2016;59(21):9645–9667.

Iwamoto N, Butler DCD, Svrzikapa N, Mohapatra S, Zlatev I, Sah DWY, et al. Control of phosphorothioate stereochemistry substantially increases the efficacy of antisense oligonucleotides. *Nat Biotechnol*. 2017;35(9):845–851.

Dimitrova, D. G. et al. (2019). RNA 2'-O-methylation (Nm) modification in human diseases. *Genes*, 10, 117.

Hill, A. C., & Hall, J. (2023). The MOE modification of RNA: Origins and widescale impact on the oligonucleotide therapeutics field. *Helvetica Chimica Acta*, 106, e202200169.

Hill, A. C., & Hall, J. (2023). The MOE modification of RNA: Origins and widescale impact on the oligonucleotide therapeutics field. *Helvetica Chimica Acta*, 106, e202200169.

Koshkin, A. A., Singh, S. K., Nielsen, P., Rajwanshi, V. K., Kumar, R., Meldgaard, M., Olsen, C. E., and Wengel, J. (1998). LNA (locked nucleic acids): Synthesis of the adenine, cytosine, guanine, 5-methylcytosine, thymine and uracil bicyclonucleoside monomers, oligomerisation, and unprecedented nucleic acid recognition. *Tetrahedron*, 54(14), 3607–3630.

Braasch, D. A., and Corey, D. R. (2001). Locked nucleic acids (LNA): Fine-tuning the recognition of DNA and RNA. *Chemistry & Biology*, 8(1), 1–7.

Elmén, J., Lindow, M., Silahatoglu, A., Bak, M., Christensen, M., Lind-Thomsen, A., Hedtjörn, M., Hansen, J. B., Hansen, H. F., Straarup, E. M., McCullagh, K., Kearney, P., Kauppinen, S., and others (2008). LNA-mediated microRNA silencing in non-human primates. *Nature*, 452, 896–899.

BenchChem Technical Guide (2025). 2'-O-MOE vs LNA: Comparative Guide to ASO Modifications.

BenchChem Technical Guide (2025). LNA vs 2'-O-Methyl RNA Performance Review.

IDT (2023). 2'-O-Methyl RNA modification overview.

Pallan, P. S., Allerson, C. R., Berdeja, A., Seth, P. P., Swayze, E. E., Prakash, T. P., and Egli, M. (2012). Structure and nuclease resistance of 2',4'-constrained 2'-O-methoxyethyl (cMOE) and 2'-O-ethyl (cEt) modified DNAs. *Chemical Communications*, 48(66), 8195–8197.

Seth, P. P., et al. (2009). Short antisense oligonucleotides with novel 2'-4' conformationally restricted nucleoside analogues show improved potency without increased toxicity. *Journal of Medicinal Chemistry*, 52(1), 10–13.

Elmén, J., et al. (2008). LNA-mediated microRNA silencing in non-human primates. *Nature*, 452, 896–899.

Burel, S. A., et al. (2016). Unique toxicity profiles of locked nucleic acid (LNA) antisense oligonucleotides. *Nucleic Acid Therapeutics*, 26(4), 199–213.

Wan, W. B., & Seth, P. P. (2016). The medicinal chemistry of therapeutic oligonucleotides. *Journal of Medicinal Chemistry*, 59(21), 9645–9667.

Seth, P. P., Vasquez, G., Allerson, C. A., Berdeja, A., Gaus, H., Kinberger, G. A., Prakash, T. P., Migawa, M. T., Bhat, B., & Swayze, E. E. (2010). Synthesis and biophysical evaluation of 2',4'-constrained 2'-O-ethyl and 2',4'-constrained 2'-O-methoxyethyl nucleic acid analogues. *Journal of Medicinal Chemistry*, 53(23), 8309–8318.

Zhao, B. S., & He, C. (2015). Pseudouridine in a new era of RNA modifications. *Cell Research*, 25, 153–154

Schlegel, M.K., Foster, D.J., Kel'in, A.V., Zlatev, I., Bisbe, A., Jayaraman, M., Lackey, J.G., Rajeev, K.G., Charisse, K., Harp, J. and Manoharan, M., 2017. Chirality dependent potency enhancement and structural impact of glycol nucleic acid modification on siRNA. *Journal of the American Chemical Society*, 139(25), pp.8537–8546.

Kurreck, J. (2003) 'Antisense technologies. Improvement through novel chemical modifications', *European Journal of Biochemistry*, 270(8), pp. 1628–1644.

Lennox, K.A. and Behlke, M.A. (2016) 'Chemical modification and design of antisense oligonucleotides', *Gene Therapy*, 23(6), pp. 428–440.

Lee, Y.C. and Lee, R.T., 2008. Interactions of oligosaccharides and glycoproteins with hepatic carbohydrate receptors. In: *Carbohydrates in Chemistry and Biology*. Wiley-VCH.

Morell, A.G. et al., 1966. Physical and chemical studies on ceruloplasmin. *Journal of Biological Chemistry*, 241, pp.3745–3749.

Egli, M. and Manoharan, M., 2023. Chemistry, structure and function of approved oligonucleotide therapeutics. *Nucleic Acids Research*, 51(6), pp.2529–2573.

Morell, A.G. et al., 1966. Physical and chemical studies on ceruloplasmin. *Journal of Biological Chemistry*, 241, pp.3745–3749.

RNA Biology & Mechanisms

Bennett CF, Swayze EE. RNA targeting therapeutics: molecular mechanisms of antisense oligonucleotides. *Annu Rev Pharmacol Toxicol*. 2010;50:259–293.

Crooke ST. Molecular mechanisms of antisense oligonucleotides. *Nucleic Acids Res*. 2017;45(1):1–14.

Lima WF, Rose JB, Nichols JG, Wu H, Migawa MT, Wyrzykiewicz TK, et al. Human RNase H1 discriminates between subtle variations in the structure of the heteroduplex substrate. *Mol Pharmacol*. 2007;71(1):83–91.

Hall AH, Wan J, Shaughnessy EE, Ramsay Shaw B, Alexander KA. RNA interference using boranophosphate siRNAs: structure–activity relationships. *Nucleic Acids Res*. 2004;32(20):5991–6000.

Crooke ST. Molecular mechanisms of antisense oligonucleotides. *Nucleic Acid Ther*. 2017;27(2):70–77.

Sang, A., Zhuo, S., Bochanis, A., Manautou, J. E., Bahal, R., Zhong, X.-b., & Rasmussen, T. P. (2024). Mechanisms of action of the US Food and Drug Administration-approved antisense oligonucleotide drugs. *BioDrugs*, 38, 511–526.

Sang, A., Zhuo, S., Bochanis, A., et al. (2024). Mechanisms of action of the US Food and Drug Administration-approved antisense oligonucleotide drugs. *BioDrugs*, 38, 511–526.

Crooke, S. T. (2017). Molecular mechanisms of antisense oligonucleotides. *Nucleic Acid Therapeutics*, 27(2), 70–77.

Bennett, C. F., & Swayze, E. E. (2010). RNA targeting therapeutics: molecular mechanisms of antisense oligonucleotides as a therapeutic platform. *Annual Review of Pharmacology and Toxicology*, 50, 259–293.

Karikó, K., Muramatsu, H., Welsh, F. A., Ludwig, J., Kato, H., Akira, S., & Weissman, D. (2008). Incorporation of pseudouridine into mRNA yields superior nonimmunogenic vectors with increased translational capacity and biological stability. *Mol Ther.* 16(11), 1833–1840.

Lin, T.-Y., Mehta, R., & Glatt, S. (2021). Pseudouridines in RNAs: switching atoms means shifting paradigms. *FEBS Letters*, 595, 2310–2322.

Eisenberg, E., & Levanon, E. Y. (2018). A-to-I RNA editing—immune protector and transcriptome diversifier. *Nat Rev Genet.*

Nishikura, K. (2010). Functions and regulation of RNA editing by ADAR deaminases. *Annu Rev Biochem*

Wahba, L., Amon, J.D., Koshland, D. and Vuica-Ross, M., 2011. RNase H and multiple RNA biogenesis factors cooperate to prevent RNA-DNA hybrids from generating genome instability. *Molecular Cell*, 44(6), pp.978–988.

Crooke, S.T. (2017) ‘Molecular mechanisms of antisense oligonucleotides’, *Nucleic Acid Therapeutics*, 27(2), pp. 70–77.

Bennett, C.F. and Swayze, E.E. (2010) ‘RNA targeting therapeutics: molecular mechanisms of antisense oligonucleotides as a therapeutic platform’, *Annual Review of Pharmacology and Toxicology*, 50, pp. 259–293.

Elbashir, S.M. et al., 2001. Duplexes of 21-nucleotide RNAs mediate RNA interference in cultured mammalian cells. *Nature*, 411, pp.494–498.

Zamore, P.D. (2006) ‘RNA interference: Big applause for silencing in Stockholm’, *Cell*, 127(6), pp. 1083–1086.

Elbashir, S.M., Harborth, J., Lendeckel, W., Yalcin, A., Weber, K. and Tuschl, T. (2001) ‘Duplexes of 21-nucleotide RNAs mediate RNA interference in cultured mammalian cells’, *Nature*, 411, pp. 494–498.

Miscellaneous / Supporting Studies

Eckstein F, Gindl H. Polyribonucleotides containing a phosphorothioate backbone. *Eur J Biochem.* 1970;13:558–564.

Stein CA, Cohen JS. Physicochemical properties of phosphorothioate oligodeoxynucleotides. *Nucleic Acids Res.* 1988;16(7):3209–3221.

Sahay G, Alakhova DY, Kabanov AV. Endocytosis of nanomedicines. *J Control Release.* 2010;145(3):182–195.

Nair JK, Willoughby JL, Chan A, et al. Multivalent N-acetylgalactosamine-conjugated siRNA localizes in hepatocytes and elicits robust RNAi-mediated gene silencing. *J Am Chem Soc.* 2014;136(49):16958–16961.

Wittrup A, Lieberman J. Knocking down disease: a progress report on siRNA therapeutics. *Nat Rev Genet.* 2015;16(9):543–552.

Dowdy SF. Overcoming cellular barriers for RNA therapeutics. *Nat Biotechnol.* 2017;35(3):222–229.

Abuchowski A, van Es T, Palczuk NC, Davis FF. Alteration of immunological properties of bovine serum albumin by covalent attachment of polyethylene glycol. *J Biol Chem.* 1977;252(11):3578–3581.

Alconcel SNS, Baas AS, Maynard HD. FDA-approved poly(ethylene glycol)–protein conjugate drugs. *Polym Chem.* 2011;2(7):1442–1448.

Keefe AD, Pai S, Ellington A. Aptamers as therapeutics. *Nat Rev Drug Discov.* 2010;9(7):537–550.

Fishbane S, Singh AK, et al. Peginesatide in patients with anemia undergoing hemodialysis. *N Engl J Med.* 2013;368:307–319.

Crooke ST, Witztum JL, Bennett CF, Baker BF. RNA-targeted therapeutics. *Cell Metab.* 2018;27(4):714–739.

Gaus HJ, Gupta R, Chappell AE, Østergaard ME, Swayze EE, Seth PP. Characterization of the interactions of chemically modified therapeutic nucleic acids with plasma proteins using fluorescence polarization. *Nucleic Acids Res.* 2019;47(2):1110–1122.

Seth PP, Vasquez G, Allerson CA, Berdeja A, Gaus H, Kinberger GA, et al. Synthesis and biophysical evaluation of 2',4'-constrained 2'-O-methoxyethyl nucleic acid analogues. *J Org Chem.* 2010;75(5):1569–1581.

Fitzgerald K, et al. A highly durable RNAi therapeutic inhibitor of PCSK9. *N Engl J Med.* 2017;376:41–51.

Nair JK, et al. Multivalent N-acetylgalactosamine-conjugated siRNA localizes in hepatocytes and elicits robust RNAi. *J Am Chem Soc.* 2014;136:16958–16961.

Martin, P. (1995). A new access to 2'-O-alkylated ribonucleosides and properties of 2'-O-alkylated oligoribonucleotides. *Helvetica Chimica Acta*, 78(2), 486–504.

Obika, S., Nanbu, D., Hari, Y., Morio, K., In, Y., Ishida, T., and Imanishi, T. (1998). Synthesis of 2'-O,4'-C-methyleneuridine and -cytidine. Novel bicyclic nucleosides having a fixed C3'-endo sugar puckering. *Tetrahedron Letters*, 39(30), 5401–5404.

Janssen, H. L. A., Reesink, H. W., Lawitz, E. J., Zeuzem, S., Rodriguez-Torres, M., Patel, K., van der Meer, A. J., Patick, A. K., Chen, A., Zhou, Y., Persson, R., King, B. D., Kauppinen, S., Levin, A. A., and Hodges, M. R. (2013). Treatment of HCV infection by targeting microRNA. *N Engl J Med.*, 368(18), 1685–1694.

BenchChem Technical Guide (2025). Binding affinity and Tm comparison of ASO chemistries.

Galindo-Murillo et al. (2024). Comparative MD analysis of ASO chemistries.

Janssen, H. L. A., et al. (2013). Treatment of HCV infection by targeting microRNA. *N Engl J Med.*, 368(18), 1685–1694.

Pallan, P. S., et al. (2012). Structural insights into the conformational properties of constrained ethyl (cEt)-modified nucleic acids. *Nucleic Acids Res.*, 40(7), 3372–3382.

Lanford, R. E., et al. (2010). Therapeutic silencing of microRNA-122 in primates with chronic hepatitis C virus infection. *Science*, 327(5962), 198–201.

Watts, J. K., & Corey, D. R. (2012). Silencing disease genes in the laboratory and the clinic. *Journal of Pathology*, 226(2), 365–379.

van der Ree, M. H., de Vree, J. M. L., Stelma, F., Willemse, S. B., van der Valk, M., Rietdijk, S., Molenkamp, R., Schinkel, J., Reesink, H. W., & van der Meer, A. J. (2017). Safety, tolerability, and antiviral effect of RG-101 in patients with chronic hepatitis C: A phase 1B, double-blind, randomized controlled trial. *Science Translational Medicine*, 9(394), eaal3896.

Neben, S. (2016–2018). RG-101 and miR-122 Antagonist Program Updates. TIDES Conference presentations.

Giacomini, K. M., Huang, S. M., Tweedie, D. J., Benet, L. Z., Brouwer, K. L. R., Chu, X., Dahlin, A., Evers, R., Fischer, V., Hillgren, K. M., et al. (2010). Membrane transporters in drug development. *Nat Rev Drug Discov.*, 9(3), 215–236.

Janssen, H. L. A., Reesink, H. W., Lawitz, E. J., Zeuzem, S., Rodriguez-Torres, M., Patel, K., van der Meer, A. J., Patick, A. K., Chen, A., Zhou, Y., et al. (2013). Treatment of HCV infection by targeting microRNA. *N Engl J Med.*, 368(18), 1685–1694.

van der Ree, M. H., de Vree, J. M. L., Stelma, F., Willemse, S., van der Valk, M., Rietdijk, S., Molenkamp, R., Schinkel, J., van Nuenen, A. C., Beuers, U., Hadi, S., Harbers, M., van der Veer, E., Liu, K., Grundy, J., Patick, A. K., Pavlicek, A., Blem, J., Huang, M., Grint, P., Neben, S., Gibson, N. W., Kootstra, N. A., & Reesink, H. W. (2017). Safety, tolerability, and antiviral effect of RG-101 in patients with chronic hepatitis C: A phase 1B, double-blind, randomised controlled trial. *The Lancet*, 389(10070), 709–717. [https://doi.org/10.1016/S0140-6736\(16\)31715-9](https://doi.org/10.1016/S0140-6736(16)31715-9)

Crooke, S. T., Witztum, J. L., Bennett, C. F., & Baker, B. F. (2018). RNA-targeted therapeutics. *Cell Metab.*, 27(4), 714–739.

Srinivasan, S., Torres, A. G., & Ribas de Pouplana, L. (2021). Inosine in Biology and Disease. *Genes*.

Hong, D. S., Kang, Y. K., Borad, M., Sachdev, J., Ejadi, S., Lim, H. Y., et al. (2020). Phase 1 study of MRX34, a liposomal miR-34a mimic, in patients with advanced solid tumours. *British Journal of Cancer*, 122, 1630–1637.

Mayo Clinic (n.d.) Inotersen (subcutaneous route) description and precautions. Available at: [HTTPS://WWW.MAYOCLINIC.ORG](https://www.mayoclinic.org) (31 March 2026).

Khorana, H.G. (2024) Har Gobind Khorana. Biographical Memoir by P.A. Sharp and U.L. RajBhandary. Washington, DC: National Academy of Sciences.

Zhang, L., Peritz, A. and Meggers, E., 2005. A simple glycol nucleic acid. *Journal of the American Chemical Society*, 127(12), pp.4174–4175.

Gane, E., Lim, Y.-S., Kim, J.B., Visvanathan, K., Sicard, E., Fedorak, R.N., Roberts, S.K., Massetto, B., Ye, X., Huang, Y., et al. (2023) 'Safety and antiviral activity of elebsiran (ALN-HBV02) in patients with chronic hepatitis B: results from a phase 1/2 study', *Journal of Hepatology*, 78(4), pp. 708–717.

Watts, J.K. and Corey, D.R. (2012) 'Silencing disease genes in the laboratory and the clinic', *Journal of Pathology*, 226(2), pp. 365–379.

Ashwell, G. and Morell, A.G., 1968. The role of surface carbohydrates in the hepatic recognition and transport of circulating glycoproteins. *Advances in Enzymology*, 41, pp.99–128.

Dowdy, S.F., 2017. Overcoming cellular barriers for RNA therapeutics. *Nature Biotechnology*, 35(3), pp.222–229.

Nair, J.K. et al., 2014. Multivalent N-acetylgalactosamine-conjugated siRNA localizes in hepatocytes and elicits robust RNAi-mediated gene silencing. *Journal of the American Chemical Society*, 136, pp.16958–16961.

Ashwell, G. and Morell, A.G., 1968. The role of surface carbohydrates in the hepatic recognition and transport of circulating glycoproteins. *Advances in Enzymology*, 41, pp.99–128.

Nair, J.K. et al., 2014. Multivalent N-acetylgalactosamine-conjugated siRNA localizes in hepatocytes and elicits robust RNAi-mediated gene silencing. *Journal of the American Chemical Society*, 136(49), pp.16958–16961.

Fire, A., Xu, S., Montgomery, M.K., Kostas, S.A., Driver, S.E. and Mello, C.C. (1998) 'Potent and specific genetic interference by double-stranded RNA in *Caenorhabditis elegans*', *Nature*, 391, pp. 806–811.

Crouch, R.J. and Dirksen, M.-L. (1982) 'Ribonucleases H', in *Nucleases*. Cold Spring Harbor Laboratory, pp. 211–241.

Thomas, M., White, R.L. and Davis, R.W. (1976) 'Hybridization of RNA to double-stranded DNA: formation of R-loops', *Proceedings of the National Academy of Sciences USA*, 73, pp. 2294–2298.

Nowotny, M., Cerritelli, S.M., Ghirlando, R., Gaidamakov, S.A., Crouch, R.J., Yang, W.(2008) *EMBO J* 27: 1172-1181 PubMed: 18337749 DOI: [HTTPS://DOI.ORG/10.1038/EMBOJ.2008.44](https://doi.org/10.1038/EMBOJ.2008.44)

Yuan, Y.R., Pei, Y., Chen, H.Y., Tuschl, T., Patel, D.J. (2006) A Potential Protein-RNA Recognition Event along the RISC-Loading Pathway from the Structure of *A. aeolicus* Argonaute with Externally Bound siRNA. (2006) *Structure* 14: 1557-1565 17027504 Search on PubMedSearch on PubMed Central DOI: [HTTPS://DOI.ORG/10.1016/J.STR.2006.08.009](https://doi.org/10.1016/j.str.2006.08.009) Primary Citation Related Structures: 2F8S, 2F8T

Alnylam Pharmaceuticals, Inc. (2019) *GIVLAARI (givosiran) prescribing information*. U.S. Food and Drug Administration. Available at: [HTTPS://WWW.FDA.GOV/DRUGSATFDA](https://www.fda.gov/drugsatfda)

Morais, P., Adachi, H. and Yu, Y.-T. (2021) The critical contribution of pseudouridine to mRNA COVID-19 vaccines. *Frontiers in Cell and Developmental Biology*, 9, 789427.

Biogen (2016) SPINRAZA (nusinersen) prescribing information. U.S. FDA.

Meijboom, K.E., Wood, M.J.A. and McClorey, G. (2017) Splice-switching therapy for spinal muscular atrophy. *Genes*, 8(6), 161.